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# TDK Project Report

## Towards Identifiable Symbolic Regression in Latent Space

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# Abstract

Data-driven hypothesis discovery is a central goal of modern science and engineering, and the increasing availability of time-series data has made dynamical-systems-based analysis relevant across many scientific domains. In this framework, the goal is not only to predict the temporal evolution of observed data, but to recover interpretable governing equations that describe the underlying data-generating process. This is challenging when high-dimensional observations arise from low-dimensional latent dynamics through an unknown nonlinear observation function. The Sparse Identification of Nonlinear Dynamics (SINDy) method provides a principled way to recover parsimonious analytical equations from data, but it assumes that the measured variables are already expressed in a coordinate system in which the dynamics are sparse. Autoencoder-based extensions of SINDy address this limitation by learning a latent representation jointly with sparse dynamics; however, these methods lack rigorous identifiability guarantees and, as shown in this work, may become unstable under realistic nonlinear mixing. Dynamics Contrastive Learning (DCL), in contrast, provides a representation-learning framework with affine identifiability guarantees, but does not by itself yield interpretable closed-form governing equations. This thesis therefore introduces DCL-SINDy, a latent-space system identification framework that combines the identifiability-oriented representation learning of DCL with the interpretability of SINDy.

To evaluate this framework, a benchmark pipeline was constructed using chaotic systems from the Dysts database, where the ground-truth latent dynamics and SINDy coefficient matrices are known. The pipeline generates latent trajectories from chaotic attractors, transforms them into observations through identity, linear, or nonlinear mixing functions, and evaluates model performance through metrics that quantify latent-space alignment, trajectory prediction, and recovery of the governing equations. The results show that DCL-SINDy reaches the expected baseline performance under simple observation regimes and becomes particularly advantageous under nonlinear mixing and high noise, where it significantly outperforms its main competitor in the literature, Autoencoder SINDy (AE-SINDy).

Overall, these findings suggest that contrastive representation learning can provide a useful route toward identifiable latent-space dynamics discovery when observations are nonlinearly mixed and noisy. This is important because dynamical systems provide a powerful framework for analysing time-series data across many scientific domains, including molecular systems and neuronal dynamics. As shown in this work, DCL-SINDy can already recover chaotic dynamics from noisy, nonlinearly mixed observations and outperforms AE-SINDy under this difficult setting. By combining the affine identifiability guarantees of Dynamics Contrastive Learning with the interpretability of SINDy, DCL-SINDy may allow human-readable analytical expressions to be recovered from complex measurements in a consistent manner. In this way, the framework represents a promising step toward automated knowledge discovery directly from time-series data.

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# List of Abbreviations

General abbreviations of the document.

| Abbreviation                      | Full expression                                                                |
|-----------------------------------|--------------------------------------------------------------------------------|
| AE-SINDy                          | Autoencoder SINDy                                                              |
| BAE-SINDy                         | Bayesian Autoencoder SINDy                                                     |
| CCA                               | Canonical correlation analysis                                                 |
| CEBRA                             | Consistent EmBeddings of high-dimensional Recordings using Auxiliary variables |
| DCL                               | Dynamics Contrastive Learning                                                  |
| DCL-SINDy                         | Dynamics Contrastive Learning SINDy                                            |
| DCL-SINDy <sub>backprop</sub>     | Backpropagation-based DCL-SINDy                                                |
| DCL-SINDy <sub>lin-reg</sub>      | Linear-regression-based DCL-SINDy                                              |
| DAE-SINDy                         | Deep Delay Autoencoder SINDy                                                   |
| DGP                               | Data-generating process                                                        |
| DS                                | Dynamical system                                                               |
| GT                                | Ground truth                                                                   |
| IC                                | Initial condition                                                              |
| iDynR <sup>2</sup>                | Identifiable dynamics $R^2$                                                    |
| iDynR <sup>2</sup> <sub>fwd</sub> | Forward identifiable dynamics $R^2$                                            |
| iDynR <sup>2</sup> <sub>bwd</sub> | Backward identifiable dynamics $R^2$                                           |
| iFroNorm                          | Identifiable (normalized) Frobenius norm                                       |
| iHam                              | Identifiable Hamming distance                                                  |
| InfoNCE                           | Information Noise-Contrastive Estimation                                       |
| LSQ                               | Least squares                                                                  |
| MCC                               | Mean correlation coefficient                                                   |
| MLP                               | Multilayer perceptron                                                          |
| MSE                               | Mean squared error                                                             |
| NCE                               | Noise-Contrastive Estimation                                                   |
| NRMSE                             | Normalized root mean squared error                                             |
| ODE                               | Ordinary differential equation                                                 |
| OLS                               | Ordinary least squares                                                         |
| RK4                               | Fourth-order Runge–Kutta method                                                |
| RMSE                              | Root mean squared error                                                        |
| SEM                               | Standard error of the mean                                                     |
| SINDy                             | Sparse Identification of Nonlinear Dynamics                                    |
| SINDy-PI                          | Sparse Identification of Nonlinear Dynamics with Parallel, Implicit dynamics   |
| SINDy-SHRED                       | SINDy with SHallow REcurrent Decoder networks                                  |
| SS <sub>res</sub>                 | Residual sum of squares                                                        |
| SS <sub>tot</sub>                 | Total sum of squares                                                           |
| STLSQ                             | Sequentially thresholded least squares                                         |
| TS                                | Time series                                                                    |
| W-SINDy                           | Weak-form Sparse Identification of Nonlinear Dynamics                          |

# Chapter 1

## Introduction

Data-driven hypothesis discovery is a central goal of modern science and engineering [1], and advanced measurement technologies now provide large amounts of time-series (TS) data in a growing number of scientific domains [2]. These data are often viewed as observations of an underlying dynamical system (DS) that generates the measured trajectories [3], [4], [5]. The DS perspective is an effective way to analyze such TS data [6] originating from a broad range of scientific domains, ranging from molecular systems [7] and neuronal dynamics [8], [9], through economics [10] and sociology [11], [12], to statistical physics [13] and cosmology [14], [15].

The general goal of DS-oriented TS analysis, then, is to infer the governing equations of the system from observations [6], that is, to perform system identification [16]. The specific goal of system identification in science, however, is not merely to obtain any mathematical description of the data that accurately predicts its temporal evolution, but rather to enable hypothesis discovery directly from observed data [17]. Therefore, we seek models that are interpretable, unique, and predictive.

One of the practical challenges in reaching this specific goal is that the dimensionality of the observed data may substantially differ from the dimensionality of the underlying data-generating process (DGP), as high-dimensional observations may lie on or near a lower-dimensional manifold [3], [18], [19] (a view that is well supported, e.g., in neuroscience [9], [20]). In addition, another difficulty arises from the high degree of nonlinearity and often chaotic behaviour present in many real-world systems [6]. Thus, in addition to interpretability, uniqueness, and accurate predictions, practical model discovery often also requires methods capable of learning a lower-dimensional representation of observed data generated by chaotic processes, further increasing the complexity of the task.

As a consequence, many existing methods lack one or more of the desired properties described above, resulting in only a partial understanding of the underlying natural phenomena. Models

that provide arbitrarily accurate predictions, such as neural networks, are generally too complex to be interpretable and often contain too many parameters to be identifiable (i.e., unique) [21], [22], [23]. Simple and parsimonious models, on the other hand, can usually recover the true underlying dynamics only if the data are recorded in the parsimonious coordinate system or if the appropriate model structure is known [24], [25].

This work therefore aims to design and benchmark, on a broad set of chaotic dynamical systems [26], a method that combines representation learning with sparse dynamical-system identification in order to obtain practical models that are predictive, parsimonious, interpretable, and identifiable (up to a known indeterminacy).

## 1.1. System Identification

System identification aims to uncover a mathematical model that describes the input-output relationship of a system [16]. In practice, this involves identifying both a model structure (i.e., the form of the governing equations) and the corresponding parameters [27], [28]. This, however, is not always possible, as different model structures or parameter values may produce identical input-output behavior. The concept of identifiability therefore addresses whether, given observed measurement data (and possibly known inputs), the parameters of a mathematical model can be uniquely determined from data by solving the associated inverse problem [29].

### 1.1.1 General Definition

Given a system operator  $\mathbf{S}$  that maps system inputs to outputs under a parameter vector  $\mathbf{p}$ , the property of (structural) identifiability is defined as follows [29]:

$$\forall(\mathbf{p}, \mathbf{p}') : \mathbf{S}[\mathbf{p}, \mathbf{u}] = \mathbf{S}[\mathbf{p}', \mathbf{u}] \implies \mathbf{p} = \mathbf{p}'. \quad (1.1)$$

That is, if two parameter vectors  $\mathbf{p}$  and  $\mathbf{p}'$  produce identical outputs under the same input  $\mathbf{u}$ , then the parameters must be equal. For a simple violation of this condition, consider the system  $y = (m_1 \cdot m_2)x$ . In this case, the parameter vector  $\mathbf{p} = [m_1, m_2]^T$  is not identifiable from measurements, since the equality  $m_1 m_2 = m'_1 m'_2$  does not imply  $m_1 = m'_1$  and  $m_2 = m'_2$ .

### 1.1.2 Identifiability of Latent Space Dynamics

Following Laiz et al. [30], the general definition in (1.1) can be made more explicit for the dynamical-systems perspective considered here. We assume that high-dimensional observed data  $\mathbf{y} \in \mathbb{R}^n$  (e.g., neural recordings) are generated by a set of latent variables  $\mathbf{x} \in \mathbb{R}^d$  according to a

continuous-time dynamical system of the form:

$$\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t), \mathbf{u}(t), t) + \boldsymbol{\varepsilon}(t); \quad \mathbf{y}(t) = \mathbf{g}(\mathbf{x}(t), \mathbf{u}(t), t) + \boldsymbol{\nu}(t), \quad (1.2)$$

where  $\mathbf{u}(t) \in \mathbb{R}^q$  is an optional input or control signal. The function  $\mathbf{f} : \mathbb{R}^d \times \mathbb{R}^q \times \mathbb{R} \rightarrow \mathbb{R}^d$  describes the latent dynamics, while the nonlinear and injective mixing function  $\mathbf{g} : \mathbb{R}^d \times \mathbb{R}^q \times \mathbb{R} \rightarrow \mathbb{R}^n$  maps latent states to the observation space. The terms  $\boldsymbol{\varepsilon}(t) \in \mathbb{R}^d$  and  $\boldsymbol{\nu}(t) \in \mathbb{R}^n$  represent process and observation noise, respectively. The goal of nonlinear system identification, then, is twofold:

1. Identify the latent space: infer the mixing function  $\mathbf{g}$ , as well as its inverse on the data manifold, the so-called *de-mixing* function or encoder  $\mathbf{h} : \mathbb{R}^n \mapsto \mathbb{R}^d$ , such that (in the noiseless case)  $\mathbf{h}(\mathbf{y}(t)) = \mathbf{h}(\mathbf{g}(\mathbf{x}(t))) = \mathbf{x}(t)$ .
2. Identify the vector field  $\mathbf{f}$  that propagates the state of the dynamical system forward in time in the low-dimensional latent space.

In practice, however, the encoder  $\mathbf{h}$  is often parameterized by a neural network, which, according to the identifiability notion introduced in 1.1.1, is generally not identifiable at the parameter level [22], [31]. Therefore, to preserve the utility of identifiability analysis as a model property, alternative notions of identifiability have been introduced.

### 1.1.3 Linear Identifiability

To analyze identifiability properties of neural networks such as the encoder  $\mathbf{h}$  (where symmetries in architecture render traditional identifiability concepts inapplicable [31]), a modified notion of identifiability, where certain indeterminacies are allowed, is generally utilized. Given observations  $\mathbf{y} \in \mathbb{R}^n$ , latent variables  $\mathbf{x} \in \mathbb{R}^d$ , a parameter vector  $\theta \in \Theta$ , and a latent variable model with the structure  $p_\theta(\mathbf{y}, \mathbf{x}) = p_\theta(\mathbf{y}|\mathbf{x})p_\theta(\mathbf{x})$ , we say that:

$$p_\theta(\mathbf{y}, \mathbf{x}) = p_\theta(\mathbf{y}|\mathbf{x})p_\theta(\mathbf{x}) \quad \text{is identifiable if} \quad \forall(\theta, \theta') : p_\theta(\mathbf{y}) = p_{\theta'}(\mathbf{y}) \quad \text{implies} \quad \theta \stackrel{L}{\sim} \theta', \quad (1.3)$$

where  $\stackrel{L}{\sim}$  is a pairwise equivalence relation on  $\Theta$  [22]. The notion of *linear identifiability*, then, as discussed in [31], means that the pairwise relation  $\stackrel{L}{\sim}$  is a linear map and two trained encoder models  $\mathbf{h}, \mathbf{h}' : \mathbb{R}^n \mapsto \mathbb{R}^d$  are identifiable if  $\mathbf{h}'(\mathbf{y}) = \mathbf{L}\mathbf{h}(\mathbf{y})$  for all relevant observations  $\mathbf{y}$ , where  $\mathbf{L} \in \text{GL}(d)$  is an invertible linear map. The complementary identifiability concepts are summarized in Box 1.1.

### Box 1.1: Summary of Identifiability Concepts

**General identifiability.** A model is identifiable if identical input-output behavior implies equality of the underlying parameters. A system operator  $S$  with parameter vector  $\mathbf{p}$  and input  $\mathbf{u}$  is said to be identifiable if

$$\forall(\mathbf{p}, \mathbf{p}') : S[\mathbf{p}, \mathbf{u}] = S[\mathbf{p}', \mathbf{u}] \implies \mathbf{p} = \mathbf{p}'.$$

**Identifiability of Encoders.** Neural-network models are identifiable only up to a known indeterminacy class  $\mathcal{L}$ . Therefore, identifiability of a latent variable model as defined in (1.3) w.r.t a parameter vector  $\theta \in \Theta$  means [22], [31]:

$$\forall(\theta, \theta') : p_\theta(\mathbf{y}) = p_{\theta'}(\mathbf{y}) \implies \theta \mathcal{L} \theta'.$$

Encoders  $\mathbf{h}, \mathbf{h}'$  are linearly identifiable if their embeddings agree up to an invertible linear transformation:

$$\forall \mathbf{y} \in \text{supp}(p_{\mathbf{y}}) : \mathbf{h}'(\mathbf{y}) = \mathbf{L}\mathbf{h}(\mathbf{y}) \in \mathbb{R}^d, \quad \mathbf{L} \in \text{GL}(d).$$

**Identifiability of Dynamics.** In the dynamical-systems setting, identifiability means that the vector field  $\mathbf{f} : \mathbb{R}^d \mapsto \mathbb{R}^d$  governing the temporal evolution of the state  $\mathbf{x} \in \mathbb{R}^d$  can be recovered from observations of the trajectory [16]:

$$\text{Given } \mathbf{x}(t) \xrightarrow{\text{infer}} \hat{\mathbf{f}} : \hat{\mathbf{f}}(\mathbf{x}(t), t) \approx \mathbf{f}(\mathbf{x}(t), t) = \dot{\mathbf{x}}(t)$$

For a parametrized model class  $\{\mathbf{f}_{\mathbf{p}}\}$ , with given governing equations, identifiability requires that

$$\forall(\mathbf{x}, t) : \mathbf{f}_{\mathbf{p}}(\mathbf{x}, t) = \mathbf{f}_{\mathbf{p}'}(\mathbf{x}, t) \implies \mathbf{p} = \mathbf{p}'.$$

**Joint Affine Identifiability in Latent Space.** Given a latent-space data-generating process  $\mathbf{x}_{t+1} = \mathbf{f}(\mathbf{x}_t) + \boldsymbol{\varepsilon}_t$  and observations  $\mathbf{y}_t = \mathbf{g}(\mathbf{x}_t) \in \mathbb{R}^n$ , the learned encoder  $\mathbf{h} : \mathbb{R}^n \mapsto \mathbb{R}^d$  and dynamics model  $\hat{\mathbf{f}} : \mathbb{R}^d \mapsto \mathbb{R}^d$  are jointly identifiable up to a shared affine transformation,  $\mathbf{L} \in \text{GL}(d)$  and  $\mathbf{b} \in \mathbb{R}^d$ , if for all predicted and true latent states  $\mathbf{x}, \hat{\mathbf{x}} \in \mathbb{R}^d$ :

1.  $\mathbf{h}$  inverts the mixing and recovers the true latents up to an affine indeterminacy:  $\mathbf{h}(\mathbf{g}(\mathbf{x})) = \hat{\mathbf{x}} = \mathbf{L}\mathbf{x} + \mathbf{b}$ .
2.  $\hat{\mathbf{f}}$  is bijective and recovers the true dynamics up to the same affine relation:  $\hat{\mathbf{f}}(\hat{\mathbf{x}}) = \mathbf{L}\mathbf{f}(\mathbf{L}^{-1}(\hat{\mathbf{x}} - \mathbf{b})) + \mathbf{b}$ .

Various methods have been developed to solve the problem of either 1 or 2, or both, from a DS perspective (see [6] for a comprehensive overview). For identifying the dynamics  $\mathbf{f}$ , a large emphasis has been placed on interpretability, leading to a surge in symbolic regression methods that yield analytical equations directly from observed data [1], [17], [24]. In parallel, several methods that invert the mixing function  $\mathbf{g}$  have been proposed with differing identifiability conditions and guarantees [30], [32], [33]. In the next sections, we introduce both pure dynamics methods as well as joint dynamics and representation-learning methods relevant to this work.

## 1.2. Relevant Computational Methods

### 1.2.1 Sparse Identification of Nonlinear Dynamics (SINDy)

Given a general nonlinear system  $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, t)$ , the task of nonlinear system identification is to infer the vector-valued function  $\mathbf{f}$  from finite observations of the state  $\mathbf{x}$  [16]. Since this task frequently occurs in fields where the main goal is knowledge discovery, rather than mere predictive power, the interpretability of the discovered models becomes a key goal [24].

The notion of interpretability here is twofold: we seek methods that can accurately predict TS data using 1) as few variables as possible while they 2) recover human-readable analytical expressions from the data. The *sparse identification of nonlinear dynamics* method developed by Brunton et al. [24] provides exactly these properties and is discussed in this section.

**Overview.** SINDy combines symbolic regression [1], a method that fits analytical expressions to data (traditionally using genetic algorithms [34]), with sparse regression [35] to identify parsimonious governing equations for a continuous-time dynamical system  $\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t), t)$ . The method constructs three data matrices, shown in Eq. (1.4):

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}^\top(t_1) \\ \mathbf{x}^\top(t_2) \\ \vdots \\ \mathbf{x}^\top(t_m) \end{bmatrix}, \quad \dot{\mathbf{X}} = \begin{bmatrix} \dot{\mathbf{x}}^\top(t_1) \\ \dot{\mathbf{x}}^\top(t_2) \\ \vdots \\ \dot{\mathbf{x}}^\top(t_m) \end{bmatrix}, \quad \Theta(\mathbf{X}) = \begin{bmatrix} | & | & | & | & & | & | \\ \mathbf{1} & \mathbf{X} & \mathbf{X}^{\mathbf{p}_2} & \mathbf{X}^{\mathbf{p}_3} & \dots & \sin(\mathbf{X}) & \cos(\mathbf{X}) & \dots \\ | & | & | & | & & | & | \end{bmatrix}$$

with  $\dot{\mathbf{x}}(t_i)^\top = \frac{d}{dt}[x_1(t_i), x_2(t_i), x_3(t_i), \dots, x_{d-2}(t_i), x_{d-1}(t_i), x_d(t_i)] \in \mathbb{R}^d \quad (1.4)$

Here, each row of  $\mathbf{X} \in \mathbb{R}^{m \times d}$  and  $\dot{\mathbf{X}} \in \mathbb{R}^{m \times d}$  corresponds to a snapshot of the latent state  $\mathbf{x} \in \mathbb{R}^d$  and its time derivative at one of  $m$  time points ( $t_i, i = 1, 2, \dots, m$ ). The matrix  $\Theta(\mathbf{X}) \in \mathbb{R}^{m \times p}$  defines a library of  $p$  candidate nonlinear functions, evaluated row-wise on the data, which serve as regressors in the sparse regression problem.

Using this library, SINDy represents the continuous-time dynamical system as  $\dot{\mathbf{X}} = \Theta(\mathbf{X})\Xi$ , where  $\Xi = [\xi_1, \xi_2, \dots, \xi_d] \in \mathbb{R}^{p \times d}$  is a sparse coefficient matrix whose columns correspond to the governing equations of the individual state variables. The goal then is to find  $\Xi$  by solving the following optimization problem:  $\hat{\Xi} = \arg \min_{\Xi} \|\Theta(\mathbf{X})\Xi - \dot{\mathbf{X}}\|_F^2 + \lambda \|\Xi\|_1$ , or an approximate sparse-regression variant of this problem, via the sequentially thresholded LSQ (STLSQ) algorithm [36]. The SINDy method is illustrated in Fig. 1.1.

**Limitations.** While SINDy works remarkably well and yields parsimonious and interpretable models across a wide range of applications [37], [38], [39], it has two major limitations: 1) the choice of library functions requires explicit domain knowledge, and 2) it requires the data to be recorded in the right coordinate system, where the dynamics are sparse [25]. As discussed by [24], the first issue may be addressed by choosing polynomial and/or sinusoidal basis functions, since in practice these functions may represent a wide range of time-series data. However, a precondition for obtaining sparse (and thus interpretable) models is that the data are either measured in the right coordinates, in which the dynamics are sparse, or that we learn a representation of the data (i.e., an encoder  $\mathbf{h}$ ), for example with a neural network, in which the dynamics are sparse.

### 1.2.2 Autoencoder SINDy

**Overview.** In [25], Champion et al. introduced autoencoder SINDy (AE-SINDy), a method that uses an encoder to map high-dimensional observations  $\mathbf{y}(t) \in \mathbb{R}^n$  to a lower-dimensional

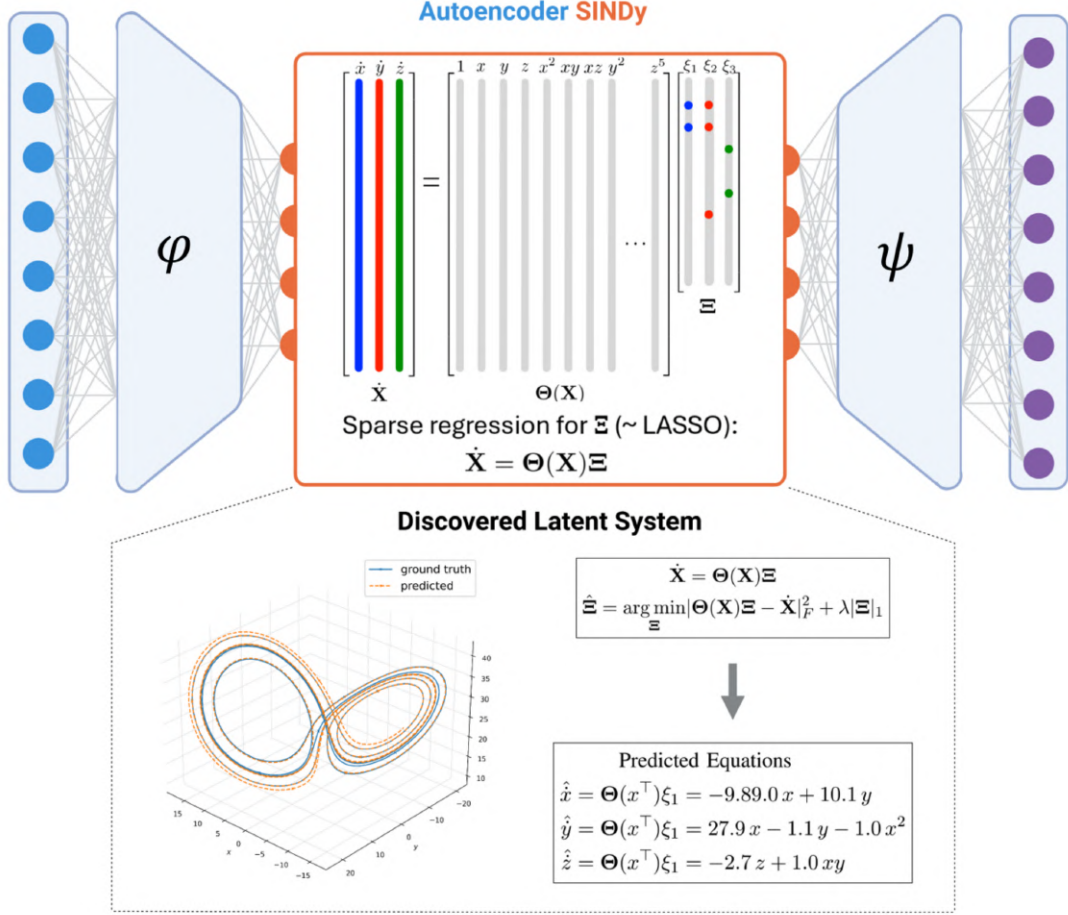


Figure 1.1: Schematic overview of SINDy and AE-SINDy. The figure summarizes both the standard SINDy workflow and its autoencoder-based extension. In SINDy, observed state trajectories are arranged into the data matrix  $\mathbf{X}$ , while the corresponding time derivatives form  $\dot{\mathbf{X}}$ . A library of candidate functions,  $\Theta(\mathbf{X})$ , is then constructed by evaluating constant, polynomial, trigonometric, or other user-defined functions on the measured states. Sparse regression is used to identify the coefficient matrix  $\Xi$ , yielding a parsimonious dynamical model of the form  $\dot{\mathbf{X}} = \Theta(\mathbf{X})\Xi$ , where each column of  $\Xi$  defines the governing equation of one state variable. AE-SINDy extends this idea to cases where the observations  $\mathbf{y}(t)$  are not recorded in the coordinate system in which the dynamics are sparse. The encoder first maps the observations to a low-dimensional latent representation  $\mathbf{x}(t)$ , SINDy identifies sparse dynamics in this latent space, and the decoder maps the latent variables back to the observation space to reconstruct the measured time series. During training, the reconstruction loss, derivative-consistency losses in both observation and latent space, and sparsity regularization on  $\Xi$  jointly encourage the learned representation to be reconstructive, dynamically consistent, and governed by sparse analytical equations. The illustration is inspired by Champion et al. [25] and the SINDy model component is borrowed from Brunton et al. [24].

latent space, fits a SINDy model on the latent variables  $\mathbf{x}(t) \in \mathbb{R}^d$ , and then maps back to the original space, reconstructing the observed TS.

Unlike SINDy, AE-SINDy learns the coefficients in the  $\xi_i$  vectors via gradient descent, jointly with the parameters of the neural network. Sparsity is enforced through sequential thresholding, that is, after every  $i^{\text{th}}$  training iteration, all entries of the coefficient matrix  $\Xi$  whose absolute

value falls below a fixed threshold are set to zero and remain fixed at 0 for the remainder of the training session. To ensure that both the learned latent dynamics and the latent representation are consistent with the observed data, AE-SINDy employs a composite loss function consisting of four terms, each targeting a different aspect of the learned representation and dynamics. Given high-dimensional observations  $\mathbf{y} \in \mathbb{R}^n$  generated from latent variables  $\mathbf{x} \in \mathbb{R}^d$  via  $\mathbf{y}(t) = \mathbf{g}(\mathbf{x}(t))$  (and denoting the decoder as  $\hat{\mathbf{g}} : \mathbb{R}^d \mapsto \mathbb{R}^n$ ), the AE-SINDy loss is defined as

$$\underbrace{\|\mathbf{y} - \hat{\mathbf{g}}(\mathbf{x})\|_2^2}_{\mathcal{L}_{\text{recon}}} + \lambda_1 \underbrace{\left\| \dot{\mathbf{y}} - (\nabla_{\mathbf{x}} \hat{\mathbf{g}}(\mathbf{x})) (\boldsymbol{\Theta}(\mathbf{x}^\top) \boldsymbol{\Xi}) \right\|_2^2}_{\mathcal{L}_{dy/dt}} + \lambda_2 \underbrace{\left\| (\nabla_{\mathbf{y}} \mathbf{x}) \dot{\mathbf{y}} - \boldsymbol{\Theta}(\mathbf{x}^\top) \boldsymbol{\Xi} \right\|_2^2}_{\mathcal{L}_{dx/dt}} + \lambda_3 \underbrace{\|\boldsymbol{\Xi}\|_1}_{\mathcal{L}_{\text{reg}}}, \quad (1.5)$$

where  $\mathcal{L}_{\text{recon}}$  is the autoencoder reconstruction loss,  $\mathcal{L}_{dy/dt}$  penalizes mismatches between the observed time derivative  $\dot{\mathbf{y}}$  and the derivative predicted via the learned latent dynamics and decoder,  $\mathcal{L}_{dx/dt}$  enforces consistency between the encoder-derived latent derivative and the SINDy model, and  $\mathcal{L}_{\text{reg}}$  promotes sparsity in the coefficient matrix  $\boldsymbol{\Xi}$ . Here,  $\nabla_{\mathbf{x}} \hat{\mathbf{g}}(\mathbf{x})$  and  $\nabla_{\mathbf{y}} \mathbf{x}$  denote the Jacobians of the decoder and encoder, respectively. Finally, the hyperparameters  $\lambda_i$  control the relative importance of the individual loss terms. The principle behind the approach is illustrated in Fig 1.1.

**Limitations.** A major limitation in Champion et al.’s work is the analysis of identifiability, both in terms of the recovered dynamics and, more substantially, in terms of the learned representations. While Champion et al. report the identification of the Lorenz system (up to an affine transform), they provide no rigorous analysis of identifiability and consistency.

Previously, Khemakhem et al. [22] showed that recovering the true joint distribution of latent variables and observations – that is, finding the true latent space with sparse dynamics for SINDy – requires explicit identifiability conditions on the model, even in the variational autoencoder framework. This raises questions about whether repeated training runs recover equivalent latent spaces and dynamics, even under non-trivial, realistic data-generating processes.

### 1.2.3 Dynamics Contrastive Learning

Contrastive learning is a representation learning approach that trains a model to bring positive (similar) samples closer together and push dissimilar (negative) samples farther apart in an embedding space [30]. Its core idea is to learn meaningful features by comparing samples to one another, rather than relying on explicit labels.

Hyvärinen et al. [40] and later Schneider et al. [32], [33], [41] demonstrated that contrastive learning based on temporal data can enable representation learning with theoretical guarantees

of linear identifiability. Later, in a related line of work, Laiz et al. [30] presented *Dynamics Contrastive Learning* (DCL) and showed that contrastive learning may also allow for nonlinear system identification. This section introduces this algorithm, together with its identifiability guarantee and limitations.

**Overview.** Laiz et al. [30] build on previous work by Schneider et al. [33], who proposed a contrastive learning approach to model time-course data. In this framework, the current observation  $\mathbf{y}_t$  is assigned as the anchor ( $\mathbf{y}$ ), the subsequent observation  $\mathbf{y}_{t+1}$  as a positive sample ( $\mathbf{y}^+$ ), and multiple observations  $\mathbf{y}_i^- \in \mathcal{N}$  (with  $i \neq t$  and  $i \neq t+1$ ) as negative samples. A contrastive loss in CEBRA [33] encourages the latent variables  $\mathbf{x}_t$  and  $\mathbf{x}_{t+1}$  corresponding to  $\mathbf{y}_t$  and  $\mathbf{y}_{t+1}$  to be close in the latent space.

DCL refines this objective by requiring not merely that the embeddings of  $\mathbf{y}_t$  and  $\mathbf{y}_{t+1}$  are close to each other, but that the prediction  $\hat{\mathbf{f}}(\mathbf{h}(\mathbf{y}_t))$  of the learned dynamics model  $\hat{\mathbf{f}}$  is close to the embedding  $\mathbf{h}(\mathbf{y}_{t+1})$ . The resulting DCL model  $\psi$  is specified by the log-likelihood given in Eq. (1.6).

$$\log p_\psi(\mathbf{y} \mid \mathbf{y}^+, \mathcal{N}) = \psi(\mathbf{y}, \mathbf{y}^+) - \log \sum_{\mathbf{y}' \in \mathcal{N} \cup \{\mathbf{y}^+\}} \exp(\psi(\mathbf{y}, \mathbf{y}')). \quad (1.6)$$

The full model  $\psi$  is trained by minimizing the negative log-likelihood over time-series data (Eq. (3) in [30]) and is parameterized to ensure favorable identifiability properties. Based on the generative model in Eq. (1.2),  $\psi$  is defined as the composition of a de-mixing function  $\mathbf{h} : \mathbb{R}^n \mapsto \mathbb{R}^d$ , an estimated dynamics model  $\hat{\mathbf{f}} : \mathbb{R}^d \mapsto \mathbb{R}^d$ , a similarity function  $\phi : \mathbb{R}^d \times \mathbb{R}^d \mapsto \mathbb{R}$ , and a correction term  $\alpha : \mathbb{R}^n \mapsto \mathbb{R}$ . Formally,

$$\psi(\mathbf{y}, \mathbf{y}') = \phi(\hat{\mathbf{f}}(\mathbf{h}(\mathbf{y})), \mathbf{h}(\mathbf{y}')) - \alpha(\mathbf{y}'). \quad (1.7)$$

The resulting algorithm takes a pair of observations  $(\mathbf{y}, \mathbf{y}')$ , maps them into a latent space via  $\mathbf{h}(\mathbf{y}) = \mathbf{L}\mathbf{x} + \mathbf{b}$  and  $\mathbf{h}(\mathbf{y}') = \mathbf{L}\mathbf{x}' + \mathbf{b}$ , applies the learned dynamics model  $\hat{\mathbf{f}}$ , and evaluates similarity using  $\phi$ . The overall procedure is illustrated in Figure 1.2a.

**Relevant Results.** Laiz et al. study a restricted, discrete-time version of the generative model in Eq. (1.2) by setting  $\mathbf{u}_t \equiv 0$  and  $\mathbf{v}_t \equiv 0$ :

$$\mathbf{x}_{t+1} = \mathbf{f}(\mathbf{x}_t) + \boldsymbol{\varepsilon}_t, \quad \mathbf{y}_t = \mathbf{g}(\mathbf{x}_t) \quad (1.8)$$

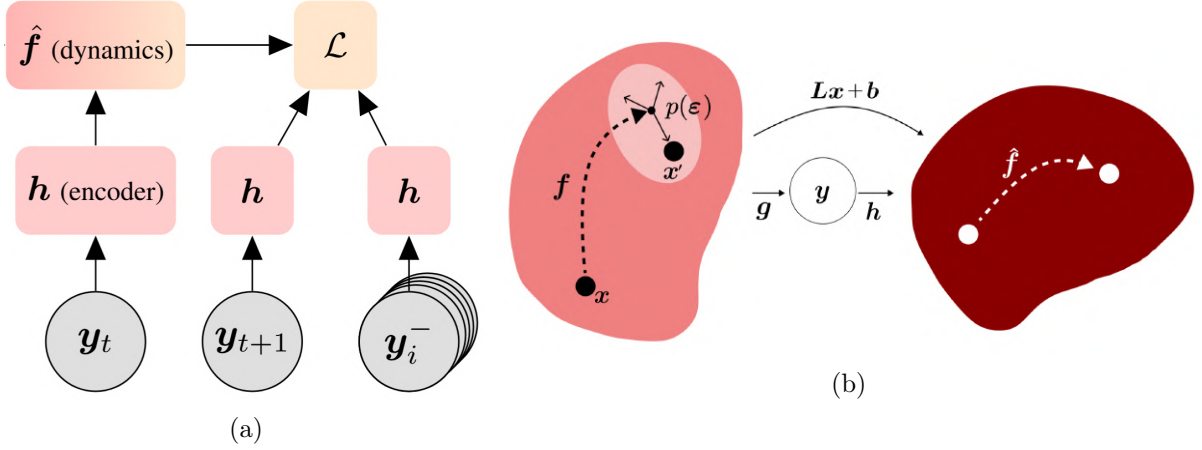


Figure 1.2: Dynamics Contrastive Learning framework and identifiability result. (a) Algorithmic structure of DCL. Pairs of observations  $(y_t, y_{t+1})$ , together with negative samples  $y_i^-$ , are mapped into a latent space by a shared de-mixing function  $h$ . A learned dynamics model  $\hat{f}$  propagates the latent representation forward in time, and the model is trained using the InfoNCE loss ( $\mathcal{L}$ ) based on a similarity measure  $\phi$ . (b) Graphical illustration of the identifiability result. Observations  $y$  are generated from ground-truth latent variables  $x$  (lighter-colored space, left) through an injective mixing function  $g$ . The learned encoder  $h$  approximately inverts the mixing and maps observations into a predicted latent space (darker-colored space, right). The true dynamics  $f$  propagate the latent state forward in time from  $x$  to  $x'$ , with additive noise  $\varepsilon$ . Under the assumptions of Theorem 1, both the estimated dynamics  $\hat{f}$  and the composition  $h \circ g$  recover the true dynamics and latent representation up to an affine transformation. The figure is borrowed from the work of Laiz et al. [30].

and provide identifiability results under additional assumptions. Specifically, Theorem 1 in [30] states that, given a time-series dataset  $\{y_t\}_{t=1}^T$  generated by a bijective dynamics function  $f : \mathbb{R}^d \mapsto \mathbb{R}^d$  and an injective mixing function  $g : \mathbb{R}^d \mapsto \mathbb{R}^n$  as defined in Eq. (1.8), with iid system noise  $\varepsilon_t \sim \mathcal{N}(0, \Sigma_\varepsilon)$ , then for the model  $\psi$ , with encoder  $h$ , estimated dynamics function  $\hat{f}$ , and similarity function  $\phi(u, v) = -\|u - v\|^2$  (that attains the global minimizer of Eq. (3) in [30]), the learned representation  $h$  and dynamics  $\hat{f}$  become identifiable (up to an affine transform) in the limit of  $T \rightarrow \infty$ .

More precisely, for any latent state  $x$  occurring with nonzero probability, the composition of the learned encoder and the true mixing function satisfies  $h(g(x)) = \hat{x} = Lx + b$ , thereby recovering the true latent variable up to an affine transformation. Furthermore, the estimated dynamics  $\hat{f}$  are bijective and obey  $\hat{f}(\hat{x}) = Lf(L^{-1}(\hat{x} - b)) + b$ , implying that  $\hat{f}$  recovers the true dynamics  $f$  up to the same affine transformation determined by  $L$ . A graphical summary of these results is shown in Fig. 1.2b.

**Limitations.** Laiz et al. evaluate their algorithm and demonstrate their theoretical identifiability results on systems exhibiting linear, switching linear, and nonlinear dynamics. However, a limitation remains. DCL approximates nonlinear dynamics using switching linear models, which

means that (i) it does not explicitly recover the true nonlinear governing equations and (ii) it does not yield an interpretable, closed-form description of the underlying dynamical system.

### 1.3. Synthesis of Methods

Since its publication by Brunton et al. in 2016 [24], the sparse identification of nonlinear dynamics algorithm has proven to be remarkably effective at yielding interpretable models from data. However, to our knowledge, one of its main practical disadvantages, namely the assumption that the data are recorded in the space where the dynamics are sparse, has previously been addressed by Champion et al. [25] in AE-SINDy, by Gao et al. [42] in the Bayesian implementation of AE-SINDy (BAE-SINDy), by Gao et al. [43] in SINDy with SHallow REcurrent Decoder networks (SINDy-SHRED), and by Bakarji et al. [44] in Deep Delay AE-SINDy (DAE-SINDy).

SINDy-SHRED and DAE-SINDy explore the case where the data-generating process lives in a higher-dimensional space than the observations, which is relevant, e.g., for spatiotemporal sensor data recorded only at a few points in space (for example, meteorological data) [43], [44]. In this work, however, we are interested in the opposite difficulty of finding a low-dimensional representation of TS data that live in a (much) higher-dimensional space, which is a major practical challenge, for example, in the field of neuroscience [9], [45], [46]. In the SINDy literature, this problem is addressed by the AE-SINDy and BAE-SINDy frameworks that aim to jointly identify a low-dimensional embedding and fit a SINDy model in this embedding space.

However, as discussed in 1.2.2, AE-SINDy lacks rigorous identifiability analysis, and therefore, in previous work (see the *Online Supplementary Material*), we investigated AE-SINDy under the data-generating process described in [30] and demonstrated that AE-SINDy and BAE-SINDy fail at representation learning when the mixing function  $\mathbf{g} : \mathbb{R}^d \mapsto \mathbb{R}^n$  is a 4-layer MLP [30], [32] instead of a simple polynomial [25].

This limitation provides the primary motivation for our current work. We aim to integrate SINDy with DCL, yielding the DCL-SINDy framework, which first uses DCL to learn an identifiable low-dimensional latent representation that is consistent across training runs [30], and then applies SINDy directly in this recovered latent space to obtain interpretable analytical expressions that identify the nonlinear dynamics within that representation up to an affine ambiguity.

# Chapter 2

## Objectives

The aim of this thesis is to develop and evaluate a latent-space system identification framework that combines the identifiability-oriented representation learning of DCL with the interpretability of SINDy. More specifically, the objectives of the work are as follows:

1. **Develop an integrated DCL-SINDy framework and a corresponding evaluation pipeline.**
  - (a) Design and implement DCL-SINDy as a framework that combines temporal contrastive representation learning with sparse identification of nonlinear dynamics.
  - (b) Construct a benchmark pipeline based on chaotic systems from an annotated database, where the ground-truth latent dynamics and SINDy coefficient matrices are known.
  - (c) Define evaluation metrics that quantify the main aspects of latent-space dynamics discovery: latent-space alignment, trajectory prediction, and recovery of the governing equations.
2. **Establish proof of concept under simpler observation regimes on the benchmark suite.**
  - (a) Test whether DCL-SINDy can recover sparse governing equations under identity mixing, where the observed variables coincide with the latent state variables.
  - (b) Extend the evaluation to linear mixing, where the latent state is observed after an injective linear transformation, and test whether DCL-SINDy can recover the latent dynamics up to the expected affine indeterminacy.
  - (c) Compare DCL-SINDy against AE-SINDy under matched data, noise, and evaluation conditions and establish that the new framework reaches baseline performance in these simpler regimes.

**3. Solve the complex problem of nonlinear system identification from nonlinearly mixed representations on a selected system from the benchmark.**

- (a) Test whether DCL-SINDy can recover the latent representation of a nonlinear chaotic system after nonlinear mixing.
- (b) Assess whether the recovered latent representation supports the discovery of the correct SINDy model in the recovered latent space.
- (c) Perform a systematic comparison between DCL-SINDy and AE-SINDy on the selected system across different nonlinear mixing depths and process-noise levels.

## Chapter 3

# Materials and Methods

The chapter provides a detailed overview of the data generation, experimentation, and model evaluation pipeline constructed for the current work. The pipeline detailed below is additionally illustrated in Fig. 3.1.

### 3.1. Data Generation

To benchmark the performance of our algorithm across a wide range of dynamics, we use the Dysts database of chaotic attractors introduced by Gilpin et al. [26], [47]. This is a highly standardized and well-annotated database of 135 chaotic systems and is available at *GilpinLab/dysts*.

Importantly for our purposes, historical parameter values of the systems known to lead to chaotic behavior are supplied within the dysts library along with known ground-truth (GT) coefficient matrices (i.e., governing equations) that can easily be compared to a fitted SINDy coefficient matrix  $\Xi$  (see Section 1.2.1). In addition, to ensure that the dynamics are consistently aligned across systems, the database contains default initial conditions and integration timesteps, selected based, for example, on ergodicity and phase surrogate significance tests, respectively [26].

Consequently, the database has been used to benchmark several system identification and forecasting algorithms ranging, for example, from foundation models [48], neural ODEs, symbolic regression [26], and state-space modeling [49], to SINDy [50]. However, to our knowledge, the database has not yet been used to benchmark latent-space system identification algorithms such as AE-SINDy or DCL that handle both nonlinear mixing and nonlinear dynamics.

In this section, we detail the integration of the dysts package into the DCL and SINDy frameworks, describe our rationale for choosing specific attractors from the database for the initial benchmark suite presented in this work, and describe the full data generation pipeline from latent dynamics to observations.

### 3.1.1 Ground Truth Latent Attractors

The implemented database comprises 71 chaotic systems in total: 57 three-dimensional systems and 14 four-dimensional hyperchaotic systems. All attractors are represented as SINDy models with a fully known ground-truth coefficient matrix  $\Xi \in \mathbb{R}^{p \times d}$ , derived from the dysts database. Therefore, the GT governing equation  $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$  is accessible as a closed-form analytical expression for each attractor:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) = (\Theta(\mathbf{x}) \Xi)^\top, \quad (3.1)$$

where  $\Theta(\mathbf{x}) \in \mathbb{R}^{1 \times p}$  is a row of polynomial library features of the state  $\mathbf{x} \in \mathbb{R}^d$ . For  $d$ -dimensional systems with polynomial order  $q$ , the library has  $p = \binom{d+q}{q}$  terms ordered by total degree in graded lexicographic order:  $\Theta(\mathbf{x}) = [1, x_1, \dots, x_d, x_1^2, x_1x_2, \dots, x_d^q]$ . For example, for  $d = 3$  and  $q = 2$ ,  $p = 10$  and a library row has the form:  $\Theta(\mathbf{x}) = [1, x, y, z, x^2, xy, xz, y^2, yz, z^2]$ .

This representation of the GT systems allows direct comparison between ground truth and discovered equations as well as between true and discovered SINDy coefficient matrices, thereby allowing for the definition of various informative metrics that we introduce in Section 3.2.3.

### 3.1.2 Generation of Observed Data

First, we generate GT trajectories from the canonical dysts initial conditions using a fourth-order Runge–Kutta (RK4) integrator with system-specific step sizes (as described in [26]). This is followed by applying a fixed, *injective* mixing function  $\mathbf{g}$  to the latent trajectory  $\mathbf{x} \in \mathbb{R}^d$ , yielding observed time-series  $\mathbf{g}(\mathbf{x}) = \mathbf{y} \in \mathbb{R}^n$ . Three mixing regimes are considered, each implemented as a frozen (non-trainable) module.

**Identity Mixing.** A simple identity transformation,  $\mathbf{g}(\mathbf{x}) = \mathbf{y} = \mathbf{x}$ , is applied, which imposes no change of coordinates, and thus the observed and latent data coincide. This scenario is the general nonlinear system identification setup that most of the previously discussed symbolic regression algorithms, such as SINDy, operate on. Therefore, in the current work, we use this mixing setup to establish that our implementation of SINDy can match the performance of the original method [24], [38].

**Linear Mixing.** A random injective linear map  $\mathbf{A} \in \mathbb{R}^{n \times d}$  is applied to map the GT data to a higher-dimensional space,  $\mathbf{g}(\mathbf{x}) = \mathbf{y} = \mathbf{A} \mathbf{x}$ . The matrix  $\mathbf{A}$  is sampled i.i.d. from  $\mathcal{N}(0, 1)$  and verified to have a trivial null space and non-zero singular values before use to ensure that the mixing is injective.

**Nonlinear Mixing.** Two nonlinear mixing variants are built from an injective MLP-based mixing map following the setup of Zimmermann et al. [32]. The network comprises  $l$  linear layers (no bias) with leaky ReLU activations (slope 0.2), and weight matrices are sampled to satisfy a condition-number threshold that enforces approximate invertibility. All parameters are frozen after initialization.

$$\mathbf{g}_{\text{nl}}(\mathbf{x}) = \mathbf{A}(\text{MLP}_l(\mathbf{x})), \quad \mathbf{g}_{\text{ln}}(\mathbf{x}) = \text{MLP}_l(\mathbf{A}\mathbf{x}). \quad (3.2)$$

The function  $\mathbf{g}_{\text{nl}} : \mathbb{R}^d \mapsto \mathbb{R}^n$  is the nonlinear composite function that first applies the MLP to the GT latent data  $\mathbf{x} \in \mathbb{R}^d$  and then applies the linear map defined by  $\mathbf{A} \in \mathbb{R}^{n \times d}$ , while  $\mathbf{g}_{\text{ln}}$  is the composite function that first maps the latent variables linearly and then applies the nonlinear mixing. In both cases,  $\mathbf{A}$  is a frozen random injective linear layer (no bias). The notation  $\mathcal{A}(\text{MLP}_l)$  is used to denote nonlinear mixing via an MLP regardless of the exact order of operations.

## 3.2. Evaluation Metrics

### 3.2.1 Latent Space Alignment

Let  $\mathbf{X} \in \mathbb{R}^{m \times d}$  denote the ground-truth latent trajectory,  $\hat{\mathbf{X}} \in \mathbb{R}^{m \times d}$  denote the recovered embedding matrix, and  $\mathbf{x} \in \mathbb{R}^d$  and  $\hat{\mathbf{x}} \in \mathbb{R}^d$  denote the ground-truth and recovered latents, respectively, at one of  $m$  time points. We use three metrics to quantify alignment between  $\mathbf{X}$  and  $\hat{\mathbf{X}}$  up to the affine indeterminacy permitted by the theoretical guarantee in 1.2.3.

**Mean Correlation Coefficient (MCC).** The MCC accounts for permutation ambiguity among latent dimensions. Given the absolute cross-correlation matrix  $\mathbf{C} \in \mathbb{R}^{d \times d}$  with entries  $C_{ij} = |\text{corr}(\hat{\mathbf{X}}_{\cdot i}, \mathbf{X}_{\cdot j})|$ , the optimal component assignment  $\sigma^* = \arg \max_{\sigma} \sum_i C_{i, \sigma(i)}$  is found via the Hungarian algorithm, and

$$\text{MCC} = \frac{1}{d} \sum_{i=1}^d C_{i, \sigma^*(i)}. \quad (3.3)$$

Pearson (default) or Spearman correlations are used for the calculation of the metric, allowing for MCC values in  $[0, 1]$ .  $\text{MCC} = 1$  indicates perfect up-to-sign and permutation linear recovery, while 0 means no correlation between the recovered dimensions (i.e., variables) and the true ones.

**Canonical Correlation Analysis (CCA).** CCA finds linear projections  $\mathbf{U}, \mathbf{V} \in \mathbb{R}^{d \times d}$  that maximally correlate the GT and recovered representations. The score is the mean of the  $d$

canonical correlations:

$$\text{CCA} = \frac{1}{d} \sum_{i=1}^d \text{corr}(\mathbf{X} \mathbf{U}_i, \hat{\mathbf{X}} \mathbf{V}_i). \quad (3.4)$$

**Linear Regression ( $R^2$ ).** To quantify affine recoverability, we compute the coefficient of determination ( $R^2$ ) in two directions. We define the *forward* direction to correspond to the prediction of the embeddings from the latents:  $\hat{\mathbf{x}} \approx \mathbf{L}_{\text{fwd}} \mathbf{x} + \mathbf{b}_{\text{fwd}}$ , and the *backward* direction to correspond to the prediction of the latents from the embeddings:  $\mathbf{x} \approx \mathbf{L}_{\text{bwd}} \hat{\mathbf{x}} + \mathbf{b}_{\text{bwd}}$ . Consequently, the coefficients of determination corresponding to the two linear regressions are

$$R_{\text{fwd}}^2 = 1 - \frac{\|\hat{\mathbf{X}} - \mathbf{X} \mathbf{L}_{\text{fwd}}^\top - \mathbf{1} \mathbf{b}_{\text{fwd}}\|_F^2}{\|\hat{\mathbf{X}} - \mathbf{1} \bar{\mathbf{x}}\|_F^2}, \quad R_{\text{bwd}}^2 = 1 - \frac{\|\mathbf{X} - \hat{\mathbf{X}} \mathbf{L}_{\text{bwd}}^\top - \mathbf{1} \mathbf{b}_{\text{bwd}}\|_F^2}{\|\mathbf{X} - \mathbf{1} \bar{\mathbf{x}}\|_F^2}, \quad (3.5)$$

where  $\mathbf{L}_{\text{fwd}}, \mathbf{L}_{\text{bwd}} \in \mathbb{R}^{d \times d}$  are the fitted linear regression matrices and  $\mathbf{b}_{\text{fwd}}, \mathbf{b}_{\text{bwd}} \in \mathbb{R}^{1 \times d}$  are the fitted intercept vectors,  $\bar{\mathbf{x}}, \bar{\hat{\mathbf{x}}} \in \mathbb{R}^{1 \times d}$  denote the temporal means of the ground-truth and recovered trajectories, respectively,  $\mathbf{1} \in \mathbb{R}^{m \times 1}$  is a vector of ones, and  $\|\cdot\|_F$  denotes the Frobenius norm. A score of  $R^2 = 1$  indicates perfect affine prediction in the corresponding direction. Intuitively, then,  $R_{\text{fwd}}^2$  of the forward fit describes how well the true latents can be mapped to the predicted embeddings, while  $R_{\text{bwd}}^2$  of the backward fit describes how well the predicted embeddings can be mapped to the true latents.

Note that for linear mixing (and trivially for identity mixing as well), the  $R^2$  score will always be 1.0 by construction of the metric, since if the mixing of the latent variables  $\mathbf{x} \in \mathbb{R}^d$  at every time point is  $\mathbf{g}(\mathbf{x}) = \mathbf{A} \mathbf{x}$ , then there will always exist an affine map for which the linear regression  $\mathbf{x} = \mathbf{L}_{\text{bwd}} \hat{\mathbf{x}} + \mathbf{b}_{\text{bwd}}$  holds. Therefore, the  $R^2$  metric is informative primarily when the mixing is nonlinear.

**Identifiable Metrics.** The  $R^2$  score as defined above is an important metric for the notion of linear (or affine) identifiability as defined in Section 1.1.3, as the  $\mathbf{L}$  matrix and  $\mathbf{b}$  vector define the affine indeterminacy up to which DCL models are identifiable ([30], Theorem 1). To account for this indeterminacy in the metrics, we report two versions of coefficient-space and trajectory-space metrics whenever the recovered latent space may differ from the ground-truth latent space. Metrics with no prefix, or with the prefix *Predictive*, quantify agreement directly in the GT latent space. Metrics with the prefix *Identifiable*, abbreviated by the leading 'i', first account for the affine transformation between the GT and recovered latent spaces. Since this affine alignment can be performed in two directions, identifiable metrics are reported in a forward direction, corresponding to  $\mathbf{x} \mapsto \hat{\mathbf{x}}$ , and in a backward direction, corresponding to  $\hat{\mathbf{x}} \mapsto \mathbf{x}$ . The concrete notation for the identifiable metrics is introduced in the corresponding sections below.

### 3.2.2 Transformation of SINDy Models

If the observed space differs from the GT latent space, then the discovered latent space will align with the true one only up to an affine transform. Therefore, as described in Section 3.2.1, to compare the recovered SINDy coefficient matrix  $\hat{\Xi}$  with the GT matrix  $\Xi$ , they first need to be transformed into the same space.

Given the DCL identifiability guarantee in Theorem 1 of [30], the true latent state  $\mathbf{x}$  and recovered embedding  $\hat{\mathbf{x}}$  satisfy the invertible affine relation  $\hat{\mathbf{x}} = \mathbf{L}\mathbf{x} + \mathbf{b}$ , with  $\mathbf{L} \in \text{GL}(d)$  and  $\mathbf{b} \in \mathbb{R}^d$ . Consequently, using the linear regression fit described in Section 3.2.1, we can transform the ground-truth coefficient matrix  $\Xi \in \mathbb{R}^{p \times d}$  into the recovered coordinate system and compare it with the fitted coefficient matrix  $\hat{\Xi} \in \mathbb{R}^{p \times d}$ .

**Problem Statement.** Given the true and predicted latent variables  $\mathbf{x}, \hat{\mathbf{x}} \in \mathbb{R}^d$  and coefficient matrices  $\Xi$  and  $\hat{\Xi}$ , the true and predicted continuous-time SINDy models of a chaotic attractor with dynamics  $\mathbf{f} : \mathbb{R}^d \mapsto \mathbb{R}^d$  are:

$$\text{true: } \frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}) = (\Theta(\mathbf{x})\Xi)^\top, \quad \text{pred: } \frac{d\hat{\mathbf{x}}}{dt} = \hat{\mathbf{f}}(\hat{\mathbf{x}}) = (\Theta(\hat{\mathbf{x}})\hat{\Xi})^\top. \quad (3.6)$$

Our aim is to transform  $\Xi$  such that the true dynamics  $\mathbf{f}$  are expressed in the recovered coordinate system, using the same polynomial library as the fitted model.

**Library Transformation.** Expressing the original latent variables in terms of the predicted ones, we get  $\mathbf{x} = \mathbf{L}^{-1}(\hat{\mathbf{x}} - \mathbf{b})$ . Consequently, any monomial  $x_{i_1}x_{i_2} \cdots x_{i_k}$ ,  $k \leq q$ , in the total-degree  $\leq q$  polynomial library row  $\Theta(\mathbf{x}) = \Theta(\mathbf{L}^{-1}(\hat{\mathbf{x}} - \mathbf{b}))$  can be expressed in terms of the predicted latents as

$$x_i = \sum_{j=1}^d (\mathbf{L}^{-1})_{ij} \hat{x}_j - (\mathbf{L}^{-1}\mathbf{b})_i = (\mathbf{L}^{-1})_{i \cdot} \hat{\mathbf{x}} - (\mathbf{L}^{-1}\mathbf{b})_i, \quad (3.7)$$

where  $(\mathbf{L}^{-1})_{i \cdot}$  is the  $i^{\text{th}}$  row of  $\mathbf{L}^{-1}$ . Therefore, the features in  $\Theta(\mathbf{x})$  can be expressed as linear combinations of the polynomial features in  $\Theta(\hat{\mathbf{x}})$ , yielding polynomials with degree  $\leq q$ . This means that there exists a matrix  $\mathbf{M}(\mathbf{L}, \mathbf{b}) \in \mathbb{R}^{p \times p}$  that performs this linear transformation on the whole feature library  $\Theta \in \mathbb{R}^{m \times p}$ , where  $m$  is the number of time points. The relationship between  $\Theta(\mathbf{x})$  and  $\Theta(\hat{\mathbf{x}})$  is:

$$\Theta(\mathbf{L}^{-1}(\hat{\mathbf{x}} - \mathbf{b}))^\top = \mathbf{M}(\mathbf{L}, \mathbf{b}) \Theta(\hat{\mathbf{x}})^\top \quad \forall \hat{\mathbf{x}}. \quad (3.8)$$

**Coefficient Transformation.** Using Eq. (3.8) and the continuous-time affine relation from [30] between the true and recovered dynamics, we get:

$$\hat{\mathbf{f}}(\hat{\mathbf{x}}) = \frac{d}{dt}(\mathbf{L}\hat{\mathbf{x}} + \mathbf{b}) = \mathbf{L}\mathbf{f}(\mathbf{x}) = \mathbf{L}(\boldsymbol{\Theta}(\mathbf{x})\boldsymbol{\Xi})^\top = \mathbf{L}\boldsymbol{\Xi}^\top \boldsymbol{\Theta}(\mathbf{L}^{-1}(\hat{\mathbf{x}} - \mathbf{b}))^\top = \mathbf{L}\boldsymbol{\Xi}^\top \mathbf{M}(\mathbf{L}, \mathbf{b})\boldsymbol{\Theta}(\hat{\mathbf{x}})^\top. \quad (3.9)$$

Finally, from Eq. (3.9) and Eq. (3.6), we get:

$$\frac{d\hat{\mathbf{x}}}{dt} = \hat{\mathbf{f}}(\hat{\mathbf{x}}) = (\boldsymbol{\Theta}(\hat{\mathbf{x}})\hat{\boldsymbol{\Xi}})^\top = \mathbf{L}\boldsymbol{\Xi}^\top \mathbf{M}(\mathbf{L}, \mathbf{b})\boldsymbol{\Theta}(\hat{\mathbf{x}})^\top \implies \boxed{\hat{\boldsymbol{\Xi}} = \mathbf{M}(\mathbf{L}, \mathbf{b})^\top \boldsymbol{\Xi} \mathbf{L}^\top}. \quad (3.10)$$

**Example.** The Lorenz system (with parameters  $\sigma = 10$ ,  $\rho = 28$ ,  $\beta = 2.667$ ) in  $d = 3$  uses the polynomial library  $\boldsymbol{\Theta}(\mathbf{x}) = [1, x, y, z, x^2, xy, xz, y^2, yz, z^2] \in \mathbb{R}^{1 \times 10}$ . With this library and given a simple linear map  $\mathbf{L} = \text{diag}(\lambda_1, \lambda_2, \lambda_3) = \text{diag}(2, 3, 5)$  ( $\mathbf{b} = 0$ ) between the GT coefficient matrix  $\boldsymbol{\Xi} \in \mathbb{R}^{10 \times 3}$  and recovered coefficient matrix  $\hat{\boldsymbol{\Xi}} \in \mathbb{R}^{10 \times 3}$ , Eq. (3.10) predicts:

$$\boldsymbol{\Xi} = \begin{bmatrix} 0 & 0 & 0 \\ -10 & 28 & 0 \\ 10 & -1 & 0 \\ 0 & 0 & -2.667 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \xRightarrow{\mathbf{L}} \quad \hat{\boldsymbol{\Xi}} = \begin{bmatrix} 0 & 0 & 0 \\ -10 & 42 & 0 \\ \frac{20}{3} & -1 & 0 \\ 0 & 0 & -2.667 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{5}{6} \\ 0 & -0.3 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (3.11)$$

and the scaling follows the per-monomial rule  $\hat{\boldsymbol{\Xi}}_{\alpha,j} = \lambda_j (\prod_{i \in \alpha} \lambda_i)^{-1} \boldsymbol{\Xi}_{\alpha,j}$ , where  $\alpha$  denotes the multi-index of the library term and  $j$  denotes the output coordinate. For example, the coefficient of the  $xz$  term in the  $\dot{y}$  equation scales as  $\lambda_2(\lambda_1\lambda_3)^{-1} = 3/(2 \cdot 5) = 0.3$ , giving  $\hat{\boldsymbol{\Xi}}_{6,1} = -0.3$ , while the coefficient of the  $xy$  term in the  $\dot{z}$  equation scales as  $\lambda_3(\lambda_1\lambda_2)^{-1} = 5/(2 \cdot 3) = 5/6$ , giving  $\hat{\boldsymbol{\Xi}}_{5,2} = 5/6$ .

### 3.2.3 Coefficient Matrix Agreement

After transforming the ground-truth SINDy model into the recovered embedding coordinates via Eq. (3.10) to obtain  $\hat{\boldsymbol{\Xi}}_{\text{pred}}$ , and aligning both libraries to a common candidate set, the agreement between  $\hat{\boldsymbol{\Xi}}_{\text{pred}}$  and  $\hat{\boldsymbol{\Xi}}$  is assessed along two axes. The structural alignment of the GT and

predicted dynamical systems is measured via the Hamming distance, while numerical agreement is quantified by matrix distance metrics.

### 3.2.3.1 Hamming Distance

We use the notion of the Hamming distance based on [51], but generalized for nonlinear dynamics via the SINDy coefficient matrices. The metric is the fraction of coefficient positions at which the support patterns *disagree*:

$$H = \frac{1}{pd} \|\mathbf{1}[\mathbf{\Xi}_{\text{true}} \neq 0] \oplus \mathbf{1}[\mathbf{\Xi}_{\text{pred}} \neq 0]\|_0, \quad (3.12)$$

where  $\oplus$  is elementwise XOR and  $\|\cdot\|_0$  counts non-zero entries. Active support is determined by an absolute threshold on coefficient magnitude.  $H = 0$  indicates identical sparsity structure;  $H = 1$  indicates complete disagreement. A *signed* variant replaces binary support with  $\text{sign}(\mathbf{\Xi}_{ij}) \cdot \mathbf{1}[\mathbf{\Xi}_{ij} \neq 0] \in \{-1, 0, +1\}$ , so sign-flips are additionally counted as disagreements. For non-identity mixing experiments, we also report the identifiable (potentially signed) Hamming distance, denoted iHam.

### 3.2.3.2 Numerical Error Metrics

Let  $\mathbf{\Xi}, \hat{\mathbf{\Xi}} \in \mathbb{R}^{p \times d}$  denote the aligned true and predicted coefficient matrices. All metrics operate on the flattened vectors  $\boldsymbol{\xi} = \text{vec}(\mathbf{\Xi})$  and  $\hat{\boldsymbol{\xi}} = \text{vec}(\hat{\mathbf{\Xi}})$ .

$$\text{MSE} = \frac{1}{pd} \sum_{i,j} (\mathbf{\Xi}_{ij} - \hat{\mathbf{\Xi}}_{ij})^2, \quad \text{RMSE} = \sqrt{\text{MSE}}, \quad \text{NRMSE} = \sqrt{\frac{\sum_{i,j} (\mathbf{\Xi}_{ij} - \hat{\mathbf{\Xi}}_{ij})^2}{\sum_{i,j} (\mathbf{\Xi}_{ij} - \bar{\mathbf{\Xi}})^2}} \quad (3.13)$$

where  $\bar{\mathbf{\Xi}}$  is the mean of all entries of  $\mathbf{\Xi}$ . Note that, by construction,  $\text{NRMSE} = \sqrt{\text{SS}_{\text{res}}/\text{SS}_{\text{tot}}} = \sqrt{1 - R_{\text{coeff}}^2}$  for the  $R^2$  score between the true and predicted coefficient matrices.

In addition to the above metrics, we use the Frobenius norm defined as

$$\|\mathbf{\Xi} - \hat{\mathbf{\Xi}}\|_F = \sqrt{\sum_{i,j} (\mathbf{\Xi}_{ij} - \hat{\mathbf{\Xi}}_{ij})^2} \quad (3.14)$$

along with its normalized variant  $\|\mathbf{\Xi} - \hat{\mathbf{\Xi}}\|_F / \|\mathbf{\Xi}\|_F$ , which provides a scale-invariant relative error and has previously been used to evaluate SINDy models on the dysts database [50]. Unlike NRMSE, this normalization uses the  $\ell_2$  energy of  $\mathbf{\Xi}$  rather than its variance, so it remains defined when all GT coefficients are equal. For non-identity mixing experiments, the corresponding identifiable (potentially normalized) Frobenius error is denoted iFroNorm.

### 3.2.4 Trajectory Prediction

Both the ground-truth model  $\mathbf{f}$  and the recovered model  $\hat{\mathbf{f}}$  are rolled forward in time  $k$  steps from the same set of initial conditions to assess the forecasting ability of the predicted SINDy model. Let  $\mathbf{x}_i^{(k)}, \hat{\mathbf{x}}_i^{(k)} \in \mathbb{R}^d$  denote the true and predicted states at horizon  $k$  for starting point  $i$ , respectively, with  $i \in \{1, \dots, N\}$ , where  $N$  is the batch size. The following numerical metrics in (3.15) and (3.16), which measure agreement between rollout states at horizon  $k$ , are used:

$$\text{MSE}_k = \frac{1}{Nd} \sum_{i,j} \left( x_{ij}^{(k)} - \hat{x}_{ij}^{(k)} \right)^2, \quad \text{RMSE}_k = \sqrt{\text{MSE}_k}, \quad \text{NRMSE}_k = \sqrt{\frac{\sum_{i,j} \left( x_{ij}^{(k)} - \hat{x}_{ij}^{(k)} \right)^2}{\sum_{i,j} \left( x_{ij}^{(k)} - \bar{x}_j^{(k)} \right)^2}}, \quad (3.15)$$

where  $x_{ij}^{(k)}$  and  $\hat{x}_{ij}^{(k)}$  denote the  $j^{\text{th}}$  coordinates of  $\mathbf{x}_i^{(k)}$  and  $\hat{\mathbf{x}}_i^{(k)}$ , respectively, and  $\bar{x}_j^{(k)} = \frac{1}{N} \sum_i x_{ij}^{(k)}$  is the per-dimension mean of the true state at horizon  $k$ . Note again that  $\text{NRMSE}_k$  is dimensionless and satisfies  $\text{NRMSE}_k = \sqrt{1 - R_k^2}$ , where  $R_k^2$  is the coefficient of determination between the GT and predicted rollout states. For non-identity mixing experiments, when the *Identifiable* version of the metrics is utilized, we refer to the trajectory alignment metrics as `iDyn<metric_name>direction` and state the rollout length explicitly for better clarity. For example, for the identifiable, forward version of the rollout  $R^2$  metric:

$$\text{iDyn}R_{\text{fwd}}^2 := 1 - \left( \sum_{i,j} \left( (\mathbf{L}_{\text{fwd}} \mathbf{x}_i^{(k)} + \mathbf{b}_{\text{fwd}})_j - \hat{x}_{ij}^{(k)} \right)^2 / \sum_{i,j} \left( \hat{x}_{ij}^{(k)} - \bar{x}_j^{(k)} \right)^2 \right). \quad (3.16)$$

When needed, the rollout horizon can be written explicitly as  $\text{iDyn}R_{k,\text{fwd}}^2$  or  $\text{iDyn}R_{k,\text{bwd}}^2$ ; otherwise, the horizon is stated in the surrounding text or figure title. The denominator is the variance of the target trajectory in the corresponding comparison space at horizon  $k$ , providing the natural no-skill baseline for forecasting at that horizon.

## 3.3. Experiments

All experiments are conducted on the same suite of twelve chaotic attractors drawn from the dysts database, namely the Lorenz, Lorenz84, SprottA, SprottB, SprottTorus, NoseHoover, Dadras, PehlivanWei, RikitakeDynamo, Rucklidge, Sakarya, and VallisElNino systems, whose governing equations are provided in the Supplementary Chapter 7. For each system, ground-truth latent trajectories are obtained by integrating the canonical dysts dynamics with an RK4 integrator and the system-specific timestep (Section 3.1.1), starting from initial conditions (ICs) sampled from a Gaussian centred at the canonical dysts fixed point with standard deviation 0.5 in each

coordinate. Unless stated otherwise, every dataset comprises 100 such trajectories of 10,000 steps each, i.e. on the order of  $10^6$  state samples per system, to which the relevant mixing function (Section 3.1.2–3.1.2) and process noise of standard deviation  $\sigma$  are applied. Each dataset is partitioned into training, validation, and test sets with a 0.7/0.15/0.15 ratio, and to control for stochasticity across the data-generating process, mixing initialization, and split, all experiments are repeated over three dataset seeds (six in the seeded nonlinear ablation). Hyperparameters are selected on the validation set, and all reported metrics are computed on the held-out test set. Figure 3.1 depicts the training and evaluation pipeline that we utilize and also visualizes the phase portraits of the twelve chaotic systems used as our benchmark.

### 3.3.1 Models

**DCL-SINDy.** We evaluate two DCL-SINDy variants, detailed in Section 1, that share the same temporal-contrastive InfoNCE objective but differ in how the SINDy coefficient matrix  $\hat{\Xi}$  is estimated. In DCL-SINDy<sub>backprop</sub>,  $\hat{\Xi}$  is optimized jointly with the encoder parameters using Adam. Sparsity is encouraged by an  $L_1$  penalty and by a sequentially updated mask  $\mathbf{S}$ , which sets coefficients with magnitude below the threshold  $\lambda_{\text{th}}$  to zero after every  $f_{\text{th}}$  iterations. The coefficients are initialized to zero, one-step forward predictions are computed with an RK4 integrator, and an optional refinement phase freezes the sparsity mask and removes the  $L_1$  penalty to fine-tune the remaining coefficients.

In DCL-SINDy<sub>lin-reg</sub>, the encoder is trained with the same contrastive objective, but  $\hat{\Xi}$  is re-estimated at each iteration by a least-squares fit between the polynomial library evaluated on the recovered latent states and the corresponding numerically estimated latent derivatives. Depending on the experiment, this fit is either unregularized or regularized with an  $L_1$  or  $L_2$  penalty. No sequential thresholding is applied during training. When used, the post-training refinement stage refits the coefficients on a larger batch of latent states.

To avoid compute-expensive hyperparameter tuning, trained DCL-SINDy<sub>lin-reg</sub> models are thresholded only once, after visualization of the coefficient matrices. This choice admittedly makes the equation-discovery procedure less automated. However, this is acceptable for the present work, because our goal is not to develop a fully automated discovery pipeline for settings where the true equations are unknown, but rather to evaluate whether the correct equations can in principle be identified from the recovered latent space.

**AE-SINDy.** AE-SINDy [25] (Section 1.2.1) is used as the primary benchmark for DCL-SINDy. Unlike DCL-SINDy, AE-SINDy learns the latent representation with an autoencoder rather than with a contrastive encoder. The two methods also differ in how the dynamics constraint is

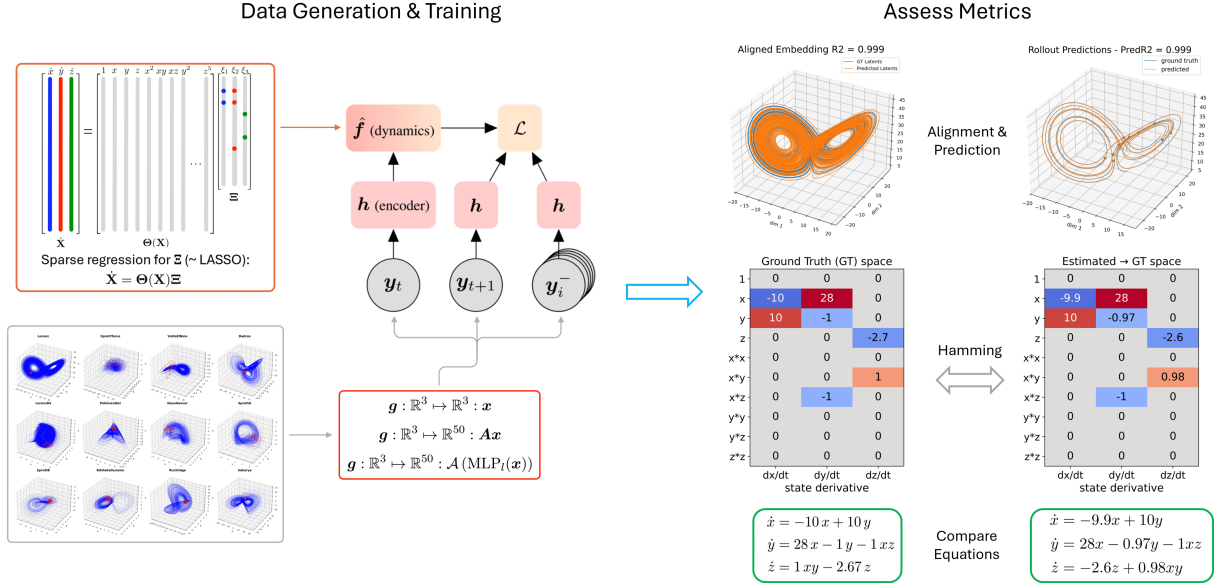


Figure 3.1: Overview of the DCL-SINDy workflow used for latent-space system identification. Ground-truth latent trajectories are first generated from chaotic attractors in the dysts benchmark suite, for which the governing equations and SINDy coefficient matrices are known. These latent states are then transformed into observations through one of three fixed mixing functions: identity mixing (Section 3.1.2), linear mixing (Section 3.1.2) into a higher-dimensional observation space, or nonlinear MLP-based mixing (Section 3.1.2). In the nonlinear experiments presented in this thesis, the main mixing function is  $\mathbf{g}_{\text{In}}(\mathbf{x}) = \text{MLP}_l(\mathbf{A}\mathbf{x})$ , i.e., the latent state is first lifted by the injective linear map  $\mathbf{A}$  and then passed through a frozen MLP. The resulting observed time series are passed to an encoder  $\mathbf{h}$ , which learns a recovered latent representation. In DCL-SINDy, the recovered states are used both for contrastive learning, through temporal positive and negative samples, and for fitting a sparse SINDy dynamics model  $\hat{\mathbf{f}}$  with coefficient matrix  $\hat{\Xi}$ . Model performance is evaluated at three levels. First, the recovered embedding is aligned with the ground-truth latent trajectory up to an affine indeterminacy allowed by DCL (Section 1.2.3), shown here by the near-perfect overlap of the aligned Lorenz attractors. Second, the learned dynamics are rolled forward from common initial conditions and compared with ground-truth rollouts. Third, the discovered SINDy coefficient matrix is transformed back into the ground-truth coordinate system and compared with the known ground-truth coefficient matrix. This enables both structural comparison through the Hamming distance and direct comparison of the recovered and ground-truth equations.

imposed. DCL-SINDy enforces the dynamics through one-step rollouts of contrastive positive pairs (Section 1.2.3), whereas AE-SINDy penalizes the discrepancy between the SINDy-predicted derivatives and finite-difference derivative estimates in both the latent and observed spaces (1.5).

The SINDy coefficient matrix is optimized jointly with the encoder and decoder parameters by gradient descent. Sparsity is imposed by sequential thresholding, where coefficients with magnitude below a fixed threshold are permanently set to zero after every  $f_{\text{th}}$  iterations. In our

experiments, the encoder and decoder are affine maps for linear mixing and three-layer MLPs of width 128 for nonlinear mixing. The coefficients are initialized to zero, one-step predictions are computed with an RK4 integrator, and training is followed by a refinement phase analogous to that used for DCL-SINDy<sub>backprop</sub>. This setup allows the two methods to be compared under matched data, mixing, and noise conditions, while isolating the effect of the representation-learning objective and the corresponding dynamics loss.

### 3.3.2 Identity Mixing

Under identity mixing, the observations coincide with the latent states,  $\mathbf{g}(\mathbf{x}) = \mathbf{y} = \mathbf{x}$ , Section 3.1.2. Therefore, no coordinate transformation has to be inverted, and the only difficulty is recovering the governing equations from noisy state observations. We use this regime as a sanity check and generate data for the following seven attractors, selected for their numerical robustness: Nose-Hoover, Sprott A, Sprott B, Dadrás, Pehlivan-Wei, Lorenz 84, and Sakarya. We consider three noise levels,  $\sigma \in \{10^{-3}, 10^{-2}, 10^{-1}\}$ , across three dataset seeds. On these data, we train DCL-SINDy<sub>backprop</sub> and AE-SINDy with matched encoders, zero-initialized coefficients, an RK4 integrator, and the same sequential-thresholding and refinement schedules, so that the comparison reflects only the difference between the learning objectives of DCL-SINDy<sub>backprop</sub> and AE-SINDy.

### 3.3.3 Linear Mixing

We next increase the difficulty by observing a linearly mixed signal,  $\mathbf{g}(\mathbf{x}) = \mathbf{y} = \mathbf{A}\mathbf{x}$ , where  $\mathbf{A} \in \mathbb{R}^{n \times d}$  is a frozen random injective map (Section 3.1.2). In this setting, the latent space and its dynamics must be recovered up to a linear transformation. Data for all twelve attractors are lifted to an observation dimension of  $n = 50$  and generated at three noise levels,  $\sigma \in \{0, 10^{-3}, 10^{-2}\}$ , across three dataset seeds, using 100 trajectories of 10,000 steps. In this regime, we train DCL-SINDy<sub>lin-reg</sub> with a linear encoder, an RK4 integrator, and a ridge least-squares solver whose regularization strength  $\alpha$  is set to zero in the noiseless case and tuned over  $\{10^{-3}, 0.05, 0.1\}$  for noisy data, optionally followed by a batch least-squares refinement. The corresponding AE-SINDy models use an affine encoder and decoder, so that their representational capacity matches the linear mixing. We sweep over the  $L_1$  weight, threshold, learning rates, and refinement length hyperparameters. As before, the best configuration is selected on the validation set, and metrics are reported on the test set, with latent-space and coefficient comparisons performed up to the affine indeterminacy described in Section 3.2.2.

### 3.3.4 Nonlinear Mixing

The most demanding regime applies an invertible nonlinear mixing constructed from a leaky-ReLU MLP,  $\mathbf{g}_{\text{in}}(\mathbf{x}) = \text{MLP}_l(\mathbf{A}\mathbf{x})$  (Section 3.1.2), whose weights are frozen after initialization under a condition-number constraint that enforces approximate invertibility. All experiments use the Lorenz system lifted to  $n = 50$  observed dimensions with 100 trajectories of 10,000 steps. Difficulty is varied along two axes simultaneously: the process-noise level,  $\sigma \in \{0, 10^{-3}, 10^{-2}\}$ , and the number of mixing layers,  $l \in \{1, 2, 3, 4\}$ , where  $l = 1$  reduces to the linear case and larger values of  $l$  produce increasingly entangled observations. To keep the comparison fair, both solvers share these data-generation settings, and the number of training iterations is set separately for each  $(l, \sigma)$  combination so that harder configurations receive a larger optimization budget. In this regime, both DCL-SINDy<sub>lin-reg</sub> and AE-SINDy use three-layer MLP architectures of width 128: DCL-SINDy<sub>lin-reg</sub> uses this architecture for its encoder, while AE-SINDy uses it for both its encoder and decoder. DCL-SINDy<sub>lin-reg</sub> is trained with a ridge regression solver with  $\alpha \in \{0, 0.05\}$ , a differentiable Savitzky–Golay derivative estimate, and an optional batch least-squares refinement, while AE-SINDy sweeps its  $L_1$  weight and threshold over a comparable grid. Each  $(l, \sigma)$  cell is repeated over six dataset seeds to reduce variance under the most unstable settings. Validation-selected models are again evaluated on the test set. Similarly to linear mixing, all alignment, coefficient, and rollout metrics are reported in their *Identifiable* form (Section 3.2.1).

## Chapter 4

# Results

### 4.1. The DCL-SINDy Algorithm

DCL-SINDy learns an encoder  $\mathbf{h}_\phi : \mathbb{R}^n \mapsto \mathbb{R}^d$  with parameters  $\phi$  to map the observed variables  $\mathbf{y} \in \mathbb{R}^n$  to recovered latent states  $\hat{\mathbf{x}} \in \mathbb{R}^d$  and estimates the true dynamics  $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, t)$  using a latent-space SINDy model  $\hat{\mathbf{f}}(\hat{\mathbf{x}}) = (\boldsymbol{\Theta}(\hat{\mathbf{x}})\hat{\boldsymbol{\Xi}})^\top$ . The training signal is a temporal-contrastive InfoNCE loss using an encoded reference  $\hat{\mathbf{x}}^r$  at time  $t$ , a positive pair  $\hat{\mathbf{x}}^+$  at  $t + \tau$ , and uniformly sampled negatives  $\hat{\mathbf{x}}^-$ , as defined in Eq. (4.1):

$$\mathcal{L}_{\text{NCE}} = \frac{1}{N} \sum_{i=1}^N \left[ \underbrace{\frac{1}{T_{\text{NCE}}} \|\tilde{\mathbf{x}}_i^+ - \hat{\mathbf{x}}_i^+\|_2^2}_{\text{alignment}} + \log \underbrace{\sum_{j=1}^M \exp \left( -\frac{1}{T_{\text{NCE}}} \|\hat{\mathbf{x}}_i^r - \hat{\mathbf{x}}_{ij}^-\|_2^2 \right)}_{\text{uniformity}} \right]. \quad (4.1)$$

Here,  $T_{\text{NCE}}$  is the InfoNCE temperature parameter, and  $\tilde{\mathbf{x}}_i^+$  is a one-step forward prediction from the current recovered state  $\hat{\mathbf{x}}_i^r$ . This is obtained by integrating the learned SINDy vector field over one time step  $\Delta t$ , such that  $\tilde{\mathbf{x}}_i^+ = \hat{\mathbf{f}}_{\mathbf{S} \odot \hat{\boldsymbol{\Xi}}}^{\Delta t}(\hat{\mathbf{x}}_i^r)$ , where  $\hat{\mathbf{f}}_{\mathbf{S} \odot \hat{\boldsymbol{\Xi}}}^{\Delta t}$  denotes the numerical flow map of the masked SINDy model with STLSQ sparsity mask  $\mathbf{S}$ . The resulting loss is thus composed of an alignment term that enforces agreement between the encoded positive sample (which is  $\hat{\mathbf{x}}$  at  $t + 1$ ) and the next state predicted by the dynamics model, and a contrastive uniformity term that separates reference states from uniformly sampled negatives. Since the same loss can be optimized with different strategies for estimating the SINDy coefficient matrix, we implemented two DCL-SINDy variants that differ only in how  $\hat{\boldsymbol{\Xi}}$  is updated. Both methods are summarized in Algorithm 1.

DCL-SINDy<sub>backprop</sub>, a backpropagation-based approach, learns the coefficient matrix jointly with the encoder weights via gradient descent, using the Adam optimizer and a sparsity mask  $\mathbf{S}$ . In this regime,  $\mathbf{S}$  defines the active support of  $\hat{\boldsymbol{\Xi}}$ :  $\mathbf{S} = \mathbb{K} \left[ |\hat{\boldsymbol{\Xi}}| \geq \lambda_{\text{th}} \right]$  for a given threshold

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**Algorithm 1** DCL-SINDy

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**Input:** observed series  $\{\mathbf{y}_t\}$ , encoder  $\mathbf{h}_\phi$ , SINDy library  $\Theta$ , step  $\Delta t$ ,  $N_{\text{iter}}$ .  
**Initialize:** encoder parameters  $\phi$ , SINDy coefficients  $\hat{\Xi} \leftarrow \mathbf{0}$ , sparsity mask  $\mathbf{S} \leftarrow \mathbf{1}$ .  
**for**  $k = 1, \dots, N_{\text{iter}}$  **do**  
  Sample contrastive batch:  $B \leftarrow (\mathbf{Y}^r, \mathbf{Y}^+, \mathbf{Y}^-)$ .  
  Encode the batch:  $\hat{\mathbf{X}}_B^r = \mathbf{h}_\phi(\mathbf{Y}^r)$ ,  $\hat{\mathbf{X}}_B^+ = \mathbf{h}_\phi(\mathbf{Y}^+)$ ,  $\hat{\mathbf{X}}_B^- = \mathbf{h}_\phi(\mathbf{Y}^-)$ .  
  **if** regression **then**  
     $\hat{\Xi} \leftarrow \arg \min_{\Xi} \left\| \Theta(\hat{\mathbf{X}}_B^r) \Xi - \mathcal{D}(\hat{\mathbf{X}}_B) \right\|_F^2 + \alpha \mathcal{R}(\Xi)$ .  
  **end if**  
  Define SINDy vector field:  $\hat{\mathbf{f}}_{\hat{\Xi}}(\hat{\mathbf{x}}) = \left( \Theta(\hat{\mathbf{x}})(\mathbf{S} \odot \hat{\Xi}) \right)^\top$ .  
  Predict positive embeddings:  $\tilde{\mathbf{X}}_B^+ \leftarrow \hat{\mathbf{f}}_{\hat{\Xi}}^{\Delta t}(\hat{\mathbf{X}}_B^r)$ .  
  Compute DCL loss:  $\mathcal{L}_{\text{DCL}} \leftarrow \mathcal{L}_{\text{NCE}}(\tilde{\mathbf{X}}_B^+, \hat{\mathbf{X}}_B^+, \hat{\mathbf{X}}_B^r, \hat{\mathbf{X}}_B^-)$ .  
  **if** not regression **then**  
     $\mathcal{L} \leftarrow \mathcal{L}_{\text{DCL}} + \lambda \left\| \mathbf{S} \odot \hat{\Xi} \right\|_1$ .  
     $(\phi, \hat{\Xi}) \leftarrow \text{Adam}(\nabla_{\phi, \hat{\Xi}} \mathcal{L})$ .  
    **if**  $k \bmod f_{\text{th}} = 0$  **then**  
       $\mathbf{S} \leftarrow \mathbf{S} \odot \mathbb{K} \left[ |\hat{\Xi}| \geq \tau_{\text{th}} \right]$ .  
       $\hat{\Xi} \leftarrow \mathbf{S} \odot \hat{\Xi}$ .  
    **end if**  
  **else**  
     $\phi \leftarrow \text{Adam}(\nabla_{\phi} \mathcal{L}_{\text{DCL}})$ .  
  **end if**  
**end for**  
**return** encoder  $\mathbf{h}_\phi$  and coefficient matrix  $\hat{\Xi}$ .

---

value  $\lambda_{\text{th}}$ . The mask  $\mathbf{S}$  is initialized as a matrix of ones and, after every  $f_{\text{th}}$  iterations, it is updated based on the current support of  $\hat{\Xi}$ . The coefficient matrix is then thresholded by the element-wise operation  $\hat{\Xi} \leftarrow \mathbf{S} \odot \hat{\Xi}$ . In addition to this thresholded algorithm, we implement a regularized least-squares-based solver (DCL-SINDy<sub>lin-reg</sub>) that learns the SINDy model by optimizing a regularized least-squares objective between the SINDy-predicted derivatives and numerical derivatives computed with a derivative estimator  $\mathcal{D}$ :

$$\hat{\Xi} \leftarrow \arg \min_{\Xi} \left\| \Theta(\hat{\mathbf{X}}_B^r) \Xi - \mathcal{D}(\hat{\mathbf{X}}_B) \right\|_F^2 + \alpha \mathcal{R}(\Xi), \quad (4.2)$$

where  $\mathcal{R}$  is a regularizer such as an L<sub>1</sub> or L<sub>2</sub> penalty. Importantly, DCL-SINDy<sub>lin-reg</sub> avoids STLSQ masking during encoder training, allowing the coefficient matrix to be re-estimated as the embedding space is reshaped by the encoder updates.

For both DCL-SINDybackprop and DCL-SINDylin-reg, post-training refinement can be performed after the main training phase. In DCL-SINDybackprop, this is done by freezing the sparsity mask  $\mathbf{S}$  and setting the L1 regularization weight  $\lambda$  to zero, while in DCL-SINDylin-reg, is achieved by refitting the LSQ objective on a larger batch using OLS, ridge, or lasso regression.

With the implemented two versions of DCL-SINDy, our aim then, is to 1) establish that DCL-SINDy achieves the baseline performance of current methods [25], [37], [38], [50] in the simple regimes of identity mixing and linear mixing and ultimately to show that DCL-SINDy significantly outperforms its main competitor, AE-SINDy, under complex, nonlinear mixing with noise, i.e., under the case that is most relevant for real-world applications.

## 4.2. DCL-SINDy Under Identity Mixing

In order to validate that both variants of DCL-SINDy work in principle, we first consider the easiest problem of recovering the dynamics under no mixing, i.e., we observe the true states  $\mathbf{x}$  directly. The performance of DCL-SINDy<sub>lin-reg</sub> is assessed on all 71 attractors of the Dysts database [26], while DCL-SINDy<sub>backprop</sub>, due to its higher computational cost, is evaluated on the  $n_{\text{attr}} = 7$  attractors selected based on their numerical robustness to noise.

We train both DCL-SINDy<sub>backprop</sub> and AE-SINDy models on the same simulation data from  $n_{\text{attr}} = 7$  attractors (Section 3.3.2) across process noise levels of  $\sigma = 0.001$ ,  $\sigma = 0.01$ , and  $\sigma = 0.1$  (see Section 3.3.2). Each attractor is simulated using  $n_{\text{seed}} = 3$  separate seeds for all noise levels, yielding 21 experiments per model per noise level. Following training, both AE-SINDy and DCL-SINDy<sub>backprop</sub> models are evaluated using the Hamming distance, the Frobenius norm, and the predictive  $R^2$  metric. Results are displayed as box plots in Fig. 4.1, with DCL-SINDy<sub>backprop</sub> models in red and AE-SINDy models in green.

In Figure 4.1a, the number of zero Hamming-distance fits is displayed across noise levels. Each point in the figure corresponds to one attractor, and the vertical axis shows the number of seeds, out of  $n_{\text{seed}} = 3$ , for which the attractor was identified with 0.0 Hamming distance by the given model. Thus, a perfect oracle model would have all points at  $y = 3$ . The modal counts of the boxes for both DCL-SINDy<sub>backprop</sub> and AE-SINDy are displayed as annotations, where  $n$  denotes the number of attractors at the corresponding seed-count level. Under the low-noise setting of  $\sigma = 0.001$ , DCL-SINDy<sub>backprop</sub> shows perfect performance, identifying all attractors across all seeds, while AE-SINDy fails to identify one attractor. As we increase the noise, however, AE-SINDy performance deteriorates substantially under  $\sigma = 0.01$  and fails completely under  $\sigma = 0.1$ . In contrast, DCL-SINDy<sub>backprop</sub> shows higher robustness to noise, identifying all attractors at least twice for  $\sigma = 0.01$  and once for  $\sigma = 0.1$ .

This trend is consistent in Fig. 4.1b, which shows the Frobenius norm of the coefficient error across noise levels for DCL-SINDy<sub>backprop</sub> (red) and AE-SINDy (green). Each point corresponds to one experiment, and the box plots summarize the distribution across the 21 experiments per model per noise level. For  $\sigma = 0.001$ , both models perform well; however, as we increase noise,

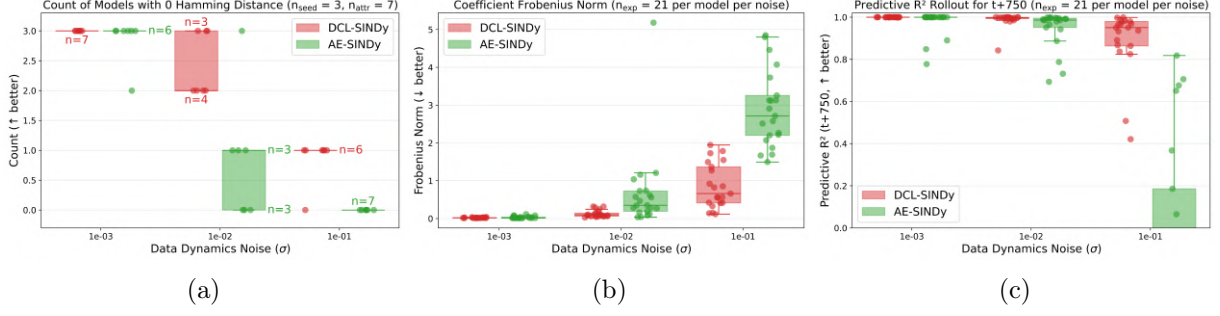


Figure 4.1: Identity-mixing benchmark of DCL-SINDy<sub>backprop</sub> and AE-SINDy across data dynamics noise levels. The box plots summarize the distributions across experiments, with boxes indicating the interquartile range, horizontal lines indicating the median, and whiskers indicating the remaining spread. DCL-SINDy<sub>backprop</sub> results are shown in red and AE-SINDy results in green. Figure 4.1a shows the count of zero Hamming-distance fits, where each point represents one attractor and the vertical axis gives the number of seeds, out of  $n_{\text{seed}} = 3$ , for which the recovered model had Hamming distance 0.0. The best score in this panel is 3, while the worst score is 0; the annotated values  $n = \dots$  indicate how many of the seven attractors lie at the corresponding seed-count level. Figure 4.1b shows the Frobenius norm of the coefficient error, where each point represents one experiment, with  $n_{\text{exp}} = 21$  experiments per model per noise level, and lower values indicate smaller coefficient error. Figure 4.1c shows the predictive  $R^2$  after a 750-step rollout, where each point represents one experiment and higher values indicate better rollout agreement with the ground-truth trajectory.

AE-SINDy performance begins to deteriorate.

Finally, we visualize 750-step rollout prediction accuracy via the predictive  $R^2$  metric in Figure 4.1c. For  $\sigma = 0.001$ , both solvers perform well; however, at  $\sigma = 0.01$ , the performance of AE-SINDy begins to deteriorate, while DCL-SINDy<sub>backprop</sub> stays practically at 1. Finally, for  $\sigma = 0.1$ , the average performance of DCL-SINDy<sub>backprop</sub> stays above 0.8, while AE-SINDy fails entirely, with 14 out of 21 models scoring below 0. In addition to Fig. 4.1c, we compute 1000-step rollout predictions for both AE-SINDy and DCL-SINDy<sub>backprop</sub> models and plot them in Supplementary Figures 7.1 and 7.2.

Figure 4.1 demonstrates that DCL-SINDy<sub>backprop</sub> can match, and under high noise even outperform, AE-SINDy, which is its direct parallel in the current literature that is capable of both interpretable dynamics identification and representation learning. Furthermore, in Supplementary Table 7.1, we evaluate the performance of DCL-SINDy<sub>lin-reg</sub> and demonstrate that, although the performance of the simple ridge regression-based model is not as robust to noise as that of DCL-SINDy<sub>backprop</sub>, it still perfectly identifies more than 50 attractors under no noise and more than 30 under  $\sigma = 0.001$  out of 71 attractors. Therefore, as a next step in complexity, in the next section, we consider linear mixing under no and low noise.

### 4.3. DCL-SINDy Solves Linear Mixing

After establishing that DCL-SINDy achieves baseline model performance under identity mixing, we next assess its performance under the more complex data-generating process of linear mixing. In this scenario, we observe  $\mathbf{y} \in \mathbb{R}^n$  and aim to identify, up to a linear transformation, both the latent space and the latent-space dynamical system  $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, t)$ . Therefore, we train  $\text{DCL-SINDy}_{\text{lin-reg}}$  models on the twelve attractors specified in Section 3.3 and shown in Supplementary Section 7.1, across three noise levels,  $\sigma = 0$ ,  $\sigma = 0.001$ , and  $\sigma = 0.01$ , and three seeds. Qualitative results for  $\sigma = 0.001$  are displayed for ten attractors in Fig. 4.2, while the results of all experiments are shown in Supplementary Figures 7.3–7.5.

Figure 4.2 displays both the forward predictions of the models for the given attractors and the discovered and true SINDy coefficient matrices. Each 1000-step rollout phase portrait shows the ground-truth test trajectory in blue, the model prediction in red, and the 75 starting points for the rollouts as red spheres. Both the forward- and backward-biased versions of the identifiable rollout  $R^2$  metric, the identifiable dynamics  $R^2$ , hereafter referred to as iDynR<sup>2</sup>, are displayed on the plots together with the name of the chaotic attractor.

Below the phase portraits, the true and discovered SINDy coefficient matrices are visualized, with rows representing the candidate functions of the  $\Theta$  feature matrix and columns representing the derivatives of the states. For example, the 2 in the GT matrix of the PehlivanWei attractor represents an offset of 2 in the differential equation for  $\frac{dz}{dt}$ . In the matrices, grey indicates zero terms, red indicates positive coefficients, and blue indicates negative coefficients. The coefficient matrices are shown in the ground-truth space, not in the discovered embedding space, and are thresholded after training at 0.7 in both the ground-truth and embedding spaces.

$\text{DCL-SINDy}_{\text{lin-reg}}$  predicts all systems with an iDynR<sup>2</sup> score above 0.99 for all but three systems, SprottTorus, SprottA, and RikitakeDynamo, while still yielding scores above 0.97 for these systems on a 1000-step horizon, which is exceptionally challenging for chaotic systems [26], [47]. In addition, as demonstrated by the alignment between the GT and predicted SINDy matrices after post-hoc thresholding (as described in Section 3.3),  $\text{DCL-SINDy}_{\text{lin-reg}}$  finds the GT equations of 9 out of the 10 chaotic attractors in Fig. 4.2, with near-perfect coefficients and failing only for the Lorenz system.

However, it is important to note that, as demonstrated in Supplementary Figures 7.3–7.5, under simple linear mixing, AE-SINDy performs just as well as  $\text{DCL-SINDy}_{\text{lin-reg}}$  for  $\sigma = 0.0$  and  $\sigma = 0.001$ , and even outperforms  $\text{DCL-SINDy}_{\text{lin-reg}}$  in most metrics when  $\sigma = 0.01$ . For our purposes, however, the more interesting question is how the two methods compare under

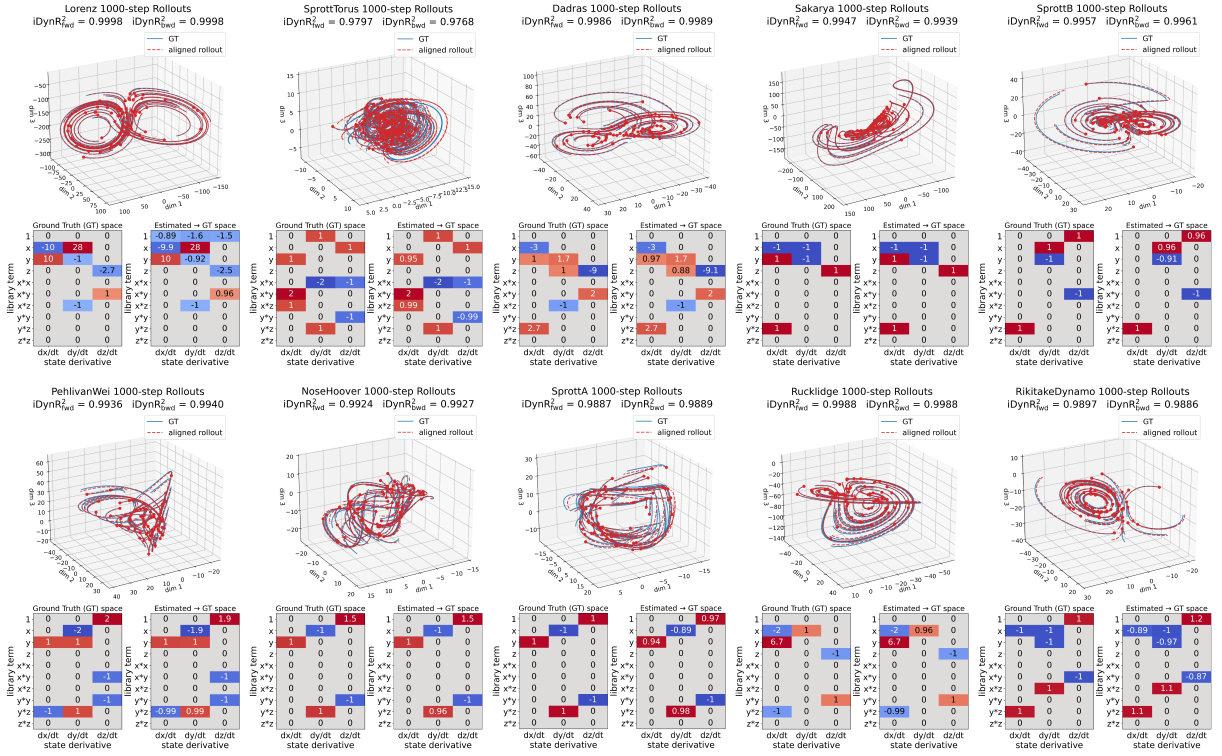


Figure 4.2: Qualitative aligned rollout and coefficient recovery results for DCL-SINDy<sub>lin-reg</sub> under linear mixing at  $\sigma = 0.001$ . The figure shows ten chaotic attractors from the linear-mixing benchmark (Section 3.3). For each attractor, the upper panel shows a 1000-step rollout phase portrait, with the ground-truth test trajectory in blue, the aligned model rollout in red, and the initial conditions marked by red spheres. The title of each phase portrait gives the attractor name and the forward- and backward-biased identifiable dynamics  $R^2$  scores, denoted as  $i\text{DynR}^2_{\text{fwd}}$  and  $i\text{DynR}^2_{\text{bwd}}$ , respectively. The lower panels show the corresponding SINDy coefficient matrices in the ground-truth coordinate system. In each matrix, rows correspond to candidate library functions in  $\Theta$  and columns correspond to the state derivatives. Grey cells indicate zero coefficients, red cells indicate positive coefficients, and blue cells indicate negative coefficients. For the shown attractors, the 1000-step rollout scores remain above 0.97, and the thresholded coefficient matrices align with the ground-truth equation structure for 9 out of 10 systems, with imperfect coefficient recovery only for the Lorenz system.

complex nonlinear mixing with high noise, which is the focus of the next section.

#### 4.4. DCL-SINDy and AE-SINDy Under Noisy Nonlinear Mixing

To study how AE-SINDy and DCL-SINDy<sub>lin-reg</sub> compare under a difficult observation regime, we consider a high-noise scenario,  $\sigma = 0.01$ , and a complex nonlinear mixing function  $\mathbf{g} : \mathbb{R}^d \rightarrow \mathbb{R}^n$ . Specifically, we use the nonlinear MLP mixing function described in Section 3.1.2:  $\mathbf{y} = \mathbf{g}_{\text{ln}}(\mathbf{x}) = \text{MLP}_l(\mathbf{A}\mathbf{x})$ . We first evaluate latent-space recovery in terms of the affine-aligned  $R^2$  score in Fig. 4.3. We then use the  $i\text{DynR}^2$  metric and rollout predictions to assess trajectory alignment in Fig. 4.4, together with the SINDy coefficient matrices to visualize qualitative agreement between the structure of the true and discovered dynamics.

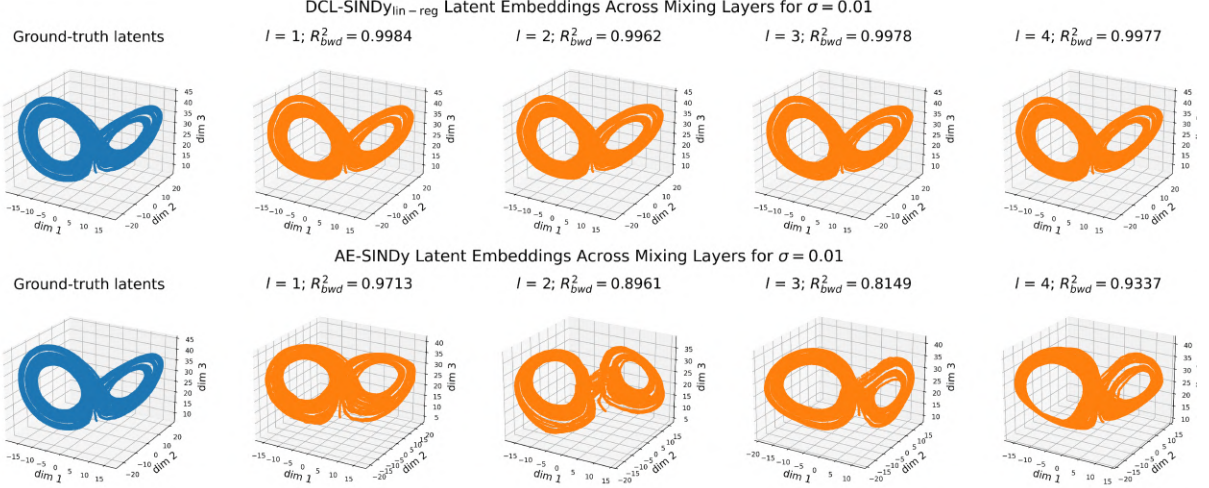


Figure 4.3: Affine-aligned latent-space recovery under noisy nonlinear mixing for DCL-SINDy<sub>lin-reg</sub> and AE-SINDy. The data are generated from the Lorenz system using the nonlinear MLP mixing function  $\mathbf{y} = \mathbf{g}_{\text{ln}}(\mathbf{x}) = \text{MLP}_l(\mathbf{A}\mathbf{x})$  at  $\sigma = 0.01$ , with  $l = 1, 2, 3, 4$  mixing layers. The top row shows DCL-SINDy<sub>lin-reg</sub> results, and the bottom row shows AE-SINDy results. In each row, the leftmost panel shows the ground-truth latent trajectory in blue, while the remaining panels show the affine-aligned recovered embeddings in orange for the corresponding number of mixing layers. The title of each recovered embedding panel gives the affine-aligned  $R^2$  score. DCL-SINDy<sub>lin-reg</sub> achieves  $R^2 > 0.99$  for all four mixing depths, while AE-SINDy reaches its highest score at  $l = 1$  with  $R^2 = 0.9713$  and remains below the DCL-SINDy<sub>lin-reg</sub> scores for all mixing depths.

In Figure 4.3, we visualize the performance of DCL-SINDy<sub>lin-reg</sub> (top row) and AE-SINDy (bottom row) in recovering the ground-truth latent space up to an affine transformation for different numbers of mixing layers,  $l = 1, 2, 3, 4$ . The  $R^2$  scores are displayed together with the affine-aligned latent embeddings in orange and the true latent trajectories in blue.

It is clear both in terms of the  $R^2$  scores and the visual alignment of the embeddings that DCL-SINDy<sub>lin-reg</sub> outperforms AE-SINDy. The DCL-SINDy<sub>lin-reg</sub>  $R^2$  scores are all above 0.99, and the embeddings align closely with the ground-truth data after affine transformation. In contrast, the best  $R^2$  score of AE-SINDy is obtained at  $l = 1$ , with  $R^2 = 0.9713$ . This decreases to  $R^2 = 0.8961$  and  $R^2 = 0.8149$  for layers 2 and 3, respectively, before rising back to  $R^2 = 0.9337$  at  $l = 4$ , possibly showing instability in the AE-SINDy models under complex mixing, a theme recurring in Section 4.5.

To investigate the second aspect of latent-space system identification, i.e., the identification of latent-space dynamics, under nonlinear mixing and a high noise of  $\sigma = 0.01$ , we visualize rollout predictions and true and recovered SINDy coefficient matrices in Fig. 4.4. The rollout predictions are shown for horizons of 250, 500, 750, and 1000 steps, computed from 75 starting points. DCL-SINDy<sub>lin-reg</sub> is shown in the first rollout row, AE-SINDy is shown in the second

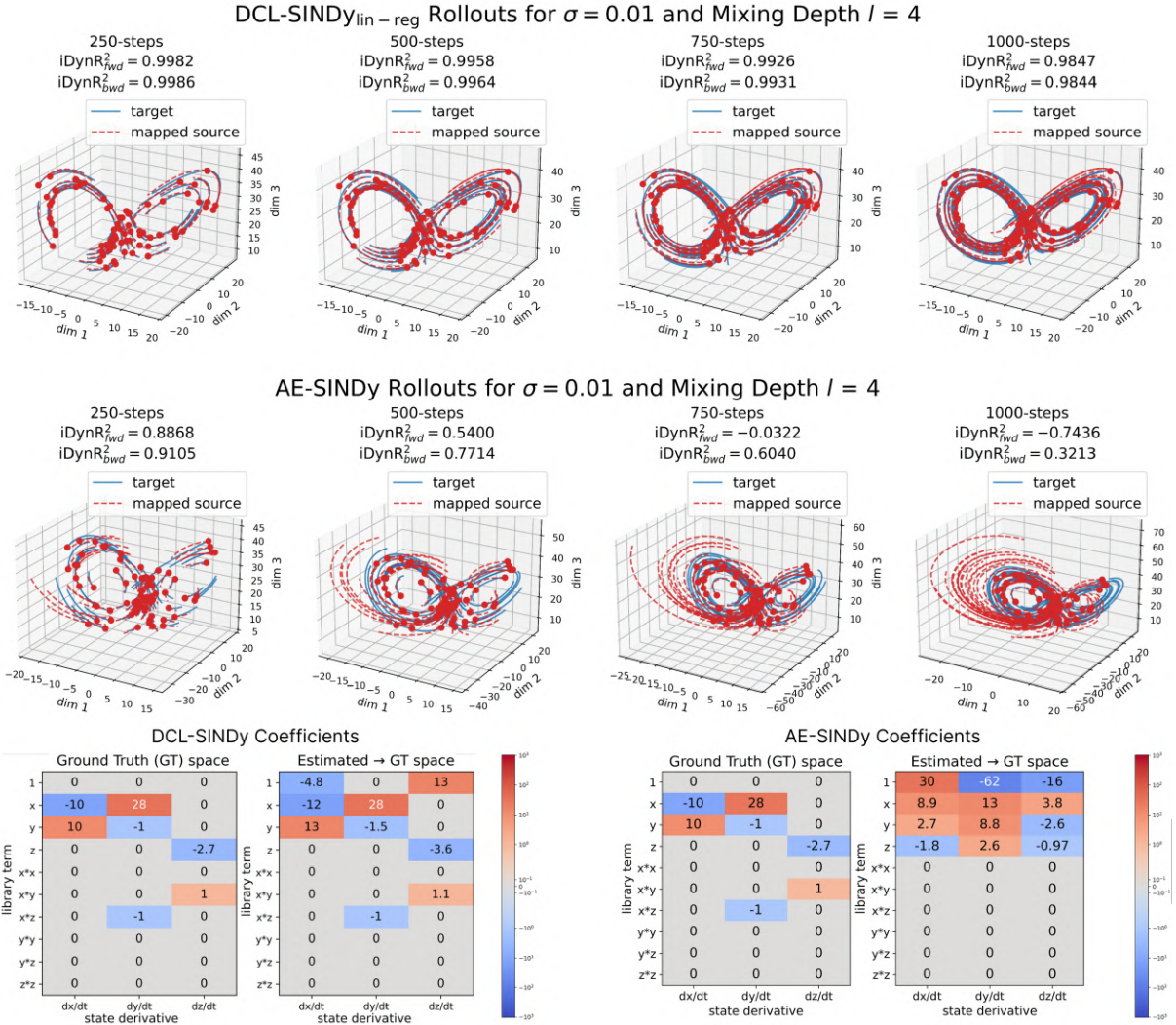


Figure 4.4: Rollout prediction and SINDy coefficient recovery for DCL-SINDy<sub>lin-reg</sub> and AE-SINDy under nonlinear Lorenz mixing with  $l = 4$  mixing layers and noise level  $\sigma = 0.01$ . The first rollout row shows DCL-SINDy<sub>lin-reg</sub> results, while the second rollout row shows AE-SINDy performance. For each model, rollout predictions are shown for horizons of 250, 500, 750, and 1000 steps. Ground-truth (GT) trajectories are shown in blue (target), aligned model predictions are shown as red dashed curves (mapped source), and rollout initial conditions are marked by red spheres. The title of each rollout panel gives the rollout horizon, while the panel annotations report the forward- and backward-biased identifiable dynamics scores for the given horizon, denoted iDynR<sub>fwd</sub><sup>2</sup> and iDynR<sub>bwd</sub><sup>2</sup>. The bottom panels show the corresponding SINDy coefficient matrices in the ground-truth coordinate system, with the DCL-SINDy<sub>lin-reg</sub> coefficients shown on the left and the AE-SINDy coefficients shown on the right. Within each coefficient block, the ground-truth matrix is shown first, followed by the estimated matrix after alignment to the ground-truth space. Rows correspond to candidate library functions and columns correspond to state derivatives. Grey cells indicate zero coefficients, red cells indicate positive coefficients, and blue cells indicate negative coefficients. DCL-SINDy<sub>lin-reg</sub> maintains iDynR<sup>2</sup> scores above 0.99 up to the 750-step horizon and remains above 0.98 at 1000 steps, while AE-SINDy rollout agreement decreases with increasing horizon and becomes negative in the forward direction at the 750- and 1000-step horizons.

rollout row, and the corresponding SINDy coefficient matrices are shown below the rollouts. The ground-truth trajectories are shown in blue, the aligned model predictions are shown in red, and the starting points for the test trajectories are displayed as red spheres. In addition to the visual alignment of predictions, the iDynR<sup>2</sup> metric is displayed for each rollout horizon in both the forward and backward directions.

From Fig. 4.4, it is apparent that DCL-SINDy<sub>lin-reg</sub> outperforms AE-SINDy under the complex regime of  $\sigma = 0.01$  and  $l = 4$ . DCL-SINDy<sub>lin-reg</sub> maintains iDynR<sup>2</sup> scores above 0.99 for all horizons up to 750 steps and remains above 0.98 even at the 1000-step horizon, which is exceptionally challenging to achieve for chaotic systems even under no mixing [26], [47]. In contrast, the iDynR<sup>2</sup> score of AE-SINDy gradually decreases as the time horizon increases and, in the forward direction, reaches negative values for rollouts of 750 steps and longer. In addition, we note that for AE-SINDy, we display validation-set alignments selected based on the best hyperparameters, as out of six trained models, we found only one that yielded finite values for iDynR<sup>2</sup> at  $l = 4$  and  $\sigma = 0.01$ . We display a model that yielded NaN values in Supplementary Figure 7.6.

The results shown in Fig. 4.4, however, illustrate the behaviour of DCL-SINDy<sub>lin-reg</sub> and AE-SINDy only for MLP depth  $l = 4$  and noise level  $\sigma = 0.01$ . Therefore, in the next section, we investigate how the performance of DCL-SINDy<sub>lin-reg</sub> and AE-SINDy compares across different noise levels and numbers of mixing layers.

## 4.5. Ablation Studies: DCL-SINDy vs. AE-SINDy

DCL-SINDy<sub>lin-reg</sub> and AE-SINDy models are trained on the Lorenz attractor across noise levels  $\sigma = 0.0$ ,  $\sigma = 0.001$ , and  $\sigma = 0.01$  and MLP mixing layer numbers  $l = 1, 2, 3, 4$ . Their performance is evaluated using the affine-aligned  $R^2$  score, the iDynR<sup>2</sup> metric, and the normalized identifiable Frobenius (iFroNorm) and signed Hamming (iHam) distances.

In Figure 4.5, the top row shows ablation plots of the affine-aligned  $R^2_{\text{bwd}}$  metric, while the bottom row displays the iDynR<sup>2</sup><sub>bwd</sub> metric across mixing layers. Each column corresponds to a different noise level, with  $\sigma = 0.0$ ,  $\sigma = 0.001$ , and  $\sigma = 0.01$  shown from left to right. The latent reconstruction metric in the top row is computed over all six trained models for each method and condition. For the rollout metric in the bottom row, however, averages are computed only over models that yield finite long-horizon rollout values; models producing NaN values during rollout evaluation are excluded from the mean and SEM. DCL-SINDy<sub>lin-reg</sub> models are represented with red lines, AE-SINDy models are shown in green, points show mean values, and whiskers show the standard error of the mean (SEM). The annotations in the bottom row mark that the AE-SINDy

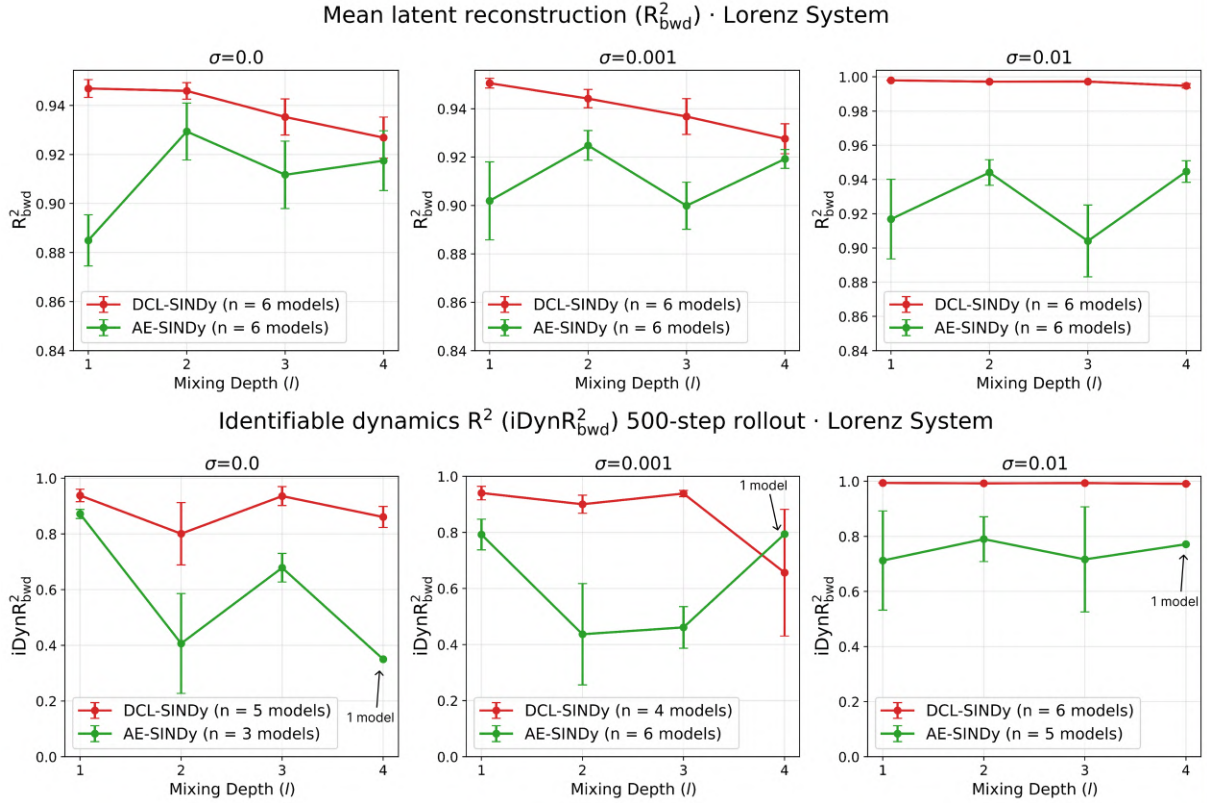


Figure 4.5: Ablation study comparing DCL-SINDy<sub>lin-reg</sub> and AE-SINDy on the Lorenz attractor under nonlinear MLP mixing. The columns correspond to noise levels  $\sigma = 0.0$ ,  $\sigma = 0.001$ , and  $\sigma = 0.01$  from left to right, and the horizontal axis in each panel gives the number of mixing layers,  $l = 1, 2, 3, 4$ . The top row shows the mean backward (i.e., estimate to GT) affine-aligned latent reconstruction score  $R^2_{\text{bwd}}$ , computed over all six trained models. The bottom row shows the backward identifiable dynamics score  $\text{iDyn}R^2_{\text{bwd}}$  for a 500-step rollout. For the rollout metric only, models that produce NaN values during long-horizon rollout evaluation are excluded from the mean and SEM, and the number of finite-rollout models is indicated in the legends. DCL-SINDy<sub>lin-reg</sub> is shown in red and AE-SINDy in green. Points show mean values and whiskers show the standard error of the mean (SEM). The annotations in the bottom row indicate that, at the deepest mixing layer, the corresponding AE-SINDy rollout point is based on only one model that was able to produce finite long-horizon rollout values. DCL-SINDy<sub>lin-reg</sub> achieves higher latent reconstruction  $R^2_{\text{bwd}}$  than AE-SINDy across all shown noise levels and mixing depths. For  $\text{iDyn}R^2_{\text{bwd}}$ , DCL-SINDy<sub>lin-reg</sub> also gives higher and more stable values in most settings, while AE-SINDy becomes less reliable at the deepest mixing layer, where only one model yields finite rollout values across the shown noise levels.

point at  $l = 4$  is based on only one finite-rollout model for each of the three noise levels.

The top row of Fig. 4.5 demonstrates that DCL-SINDy<sub>lin-reg</sub> outperforms AE-SINDy in representation learning, as measured by the affine-aligned  $R^2_{\text{bwd}}$  metric, across all mixing layer numbers and noise levels. Similarly, in the bottom row, DCL-SINDy<sub>lin-reg</sub> achieves higher and more stable  $\text{iDyn}R^2_{\text{bwd}}$  values for most mixing depths and noise levels. AE-SINDy is particularly unstable at the deepest mixing layer, where only one trained model yields finite long-horizon rollout values at  $l = 4$  for each of the three noise levels.

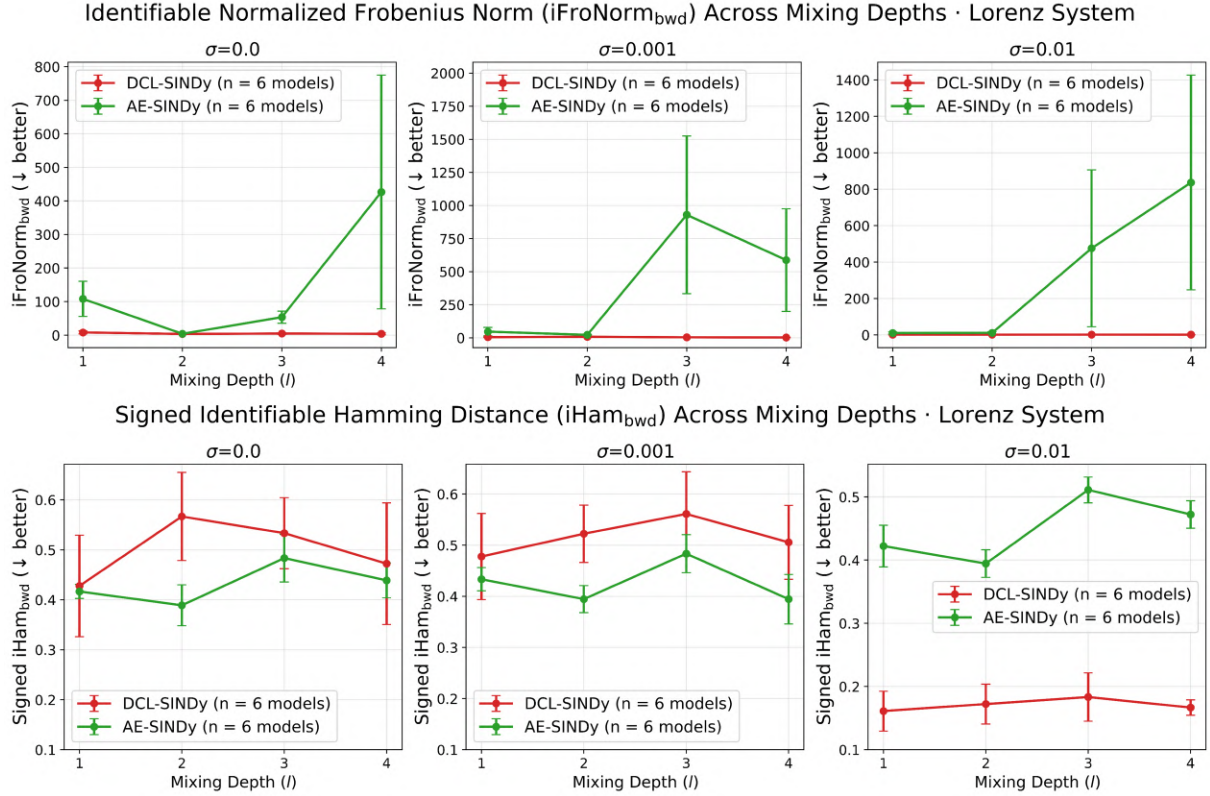


Figure 4.6: Ablation study comparing DCL-SINDy<sub>lin-reg</sub> and AE-SINDy on the Lorenz attractor using coefficient-space recovery metrics under nonlinear MLP mixing. The columns correspond to noise levels  $\sigma = 0.0$ ,  $\sigma = 0.001$ , and  $\sigma = 0.01$  from left to right, and the horizontal axis in each panel gives the number of mixing layers,  $l = 1, 2, 3, 4$ . The top row shows the backward normalized identifiable Frobenius distance, iFroNorm<sub>bwd</sub>, and the bottom row shows the backward signed identifiable Hamming distance, iHam<sub>bwd</sub>. Both metrics are computed after transforming the discovered SINDy coefficient matrix back to the ground-truth coordinate system. DCL-SINDy<sub>lin-reg</sub> is shown in red and AE-SINDy in green. Points show mean values across all six trained models, and whiskers show the standard error of the mean (SEM). Lower values indicate better coefficient recovery. DCL-SINDy<sub>lin-reg</sub> achieves lower normalized iFroNorm<sub>bwd</sub> values than AE-SINDy across all shown noise levels and mixing depths, with the largest differences at  $\sigma = 0.001$  and  $\sigma = 0.01$  for deeper MLP mixing. For signed iHam<sub>bwd</sub>, AE-SINDy gives lower values under no and low noise, while DCL-SINDy<sub>lin-reg</sub> gives lower values at  $\sigma = 0.01$ .

To investigate the qualitative reconstruction of the system dynamics, we compute and visualize the normalized iFroNorm<sub>bwd</sub> and signed iHam<sub>bwd</sub> metrics in Fig. 4.6. DCL-SINDy<sub>lin-reg</sub> models are again depicted in red, while AE-SINDy models are shown in green. In contrast to the rollout metric in Fig. 4.5, these coefficient-space metrics are computed over all six trained models for each method and condition. Points show mean values across these six models, and whiskers show the standard error of the mean (SEM). The top row shows the normalized iFroNorm<sub>bwd</sub> metric, while the bottom row depicts the signed iHam<sub>bwd</sub> metric. In both cases, lower scores indicate a smaller distance from the true coefficient matrix and therefore better recovery of the true dynamics (see Section 3.2.3 for details). Both metrics are shown in the backward direction,

i.e., after transforming the discovered SINDy matrix back to the ground-truth space.

DCL-SINDy<sub>lin-reg</sub> outperforms AE-SINDy in terms of the normalized iFroNorm<sub>bwd</sub> metric for all mixing layers and noise levels, especially for  $\sigma = 0.001$  or  $\sigma = 0.01$  and  $l = 3$  or  $l = 4$ , where the AE-SINDy coefficient errors become very large. This agrees with the rollout results in Fig. 4.4, where only one AE-SINDy model yields finite long-horizon rollout values at  $l = 4$  and  $\sigma = 0.01$ . For the signed iHam<sub>bwd</sub> metric, however, AE-SINDy achieves lower, and therefore better, scores for all mixing layers under no or low noise. This changes under  $\sigma = 0.01$ , where the signed iHam<sub>bwd</sub> distances for DCL-SINDy<sub>lin-reg</sub> decrease, while those of AE-SINDy increase.

This improvement in the performance and apparent stability of DCL-SINDy<sub>lin-reg</sub> at higher noise levels is observable across the ablation figures in Fig. 4.5 and Fig. 4.6. We explore this apparent contradiction, namely the improved performance of our model under a higher-noise regime, in the Discussion (Chapter 5) and hypothesize that it is inherent to InfoNCE-style contrastive objectives.

# Chapter 5

## Discussion

### 5.1. Combining Dynamics Contrastive Learning and SINDy

In this work, we combine Dynamics Contrastive Learning [30] with the SINDy method [24] and introduce DCL-SINDy, an algorithm that aims to recover low-dimensional, sparse representations of dynamical systems from high-dimensional observations. Unlike existing methods that concentrate either on identifying learned representations [22], [31] or on interpretable dynamics discovery [1], [17], [24], our method aims to combine both objectives and jointly identify the latent representation and the governing dynamics.

DCL-SINDy maps observations into a recovered latent space with an encoder and fits a SINDy model directly in this space. The training objective is based on temporal contrastive learning: a reference observation is encoded, propagated forward by the learned SINDy vector field, and then compared to the encoded positive sample, while negative samples are pushed away by the InfoNCE objective. This couples representation learning to the learned dynamics, rather than treating the two problems independently. Within this framework, we implement two variants. In DCL-SINDy<sub>backprop</sub>, the SINDy coefficient matrix is optimized jointly with the encoder parameters by gradient descent, with sparsity imposed through an  $L_1$  penalty and sequential thresholding. In DCL-SINDy<sub>lin-reg</sub>, the encoder is trained with the same contrastive objective, but the SINDy coefficients are re-estimated by a regularized least-squares fit during training. These two variants allow us to test both a fully differentiable dynamics-constrained objective and a simpler regression-based fitting strategy within the same DCL-SINDy framework.

With this algorithm in hand, we first asked whether DCL-SINDy reaches the baseline performance expected from existing sparse dynamics-discovery methods under simple observation regimes. We then moved to progressively harder settings, where the latent dynamics are observed only after linear or nonlinear mixing, and compared DCL-SINDy against AE-SINDy as the closest

existing method that combines representation learning with sparse dynamics discovery.

## 5.2. DCL-SINDy Outperforms Existing Methods Under Various Mixing Setups

### 5.2.1 Identity Mixing and Linear Mixing

To establish that DCL-SINDy can match existing methods, we first test it in two comparatively simple settings: identity mixing, where the observations have the same dimensionality as the dynamical system, and linear mixing, where the latent states are observed after an injective linear transformation. Both experiments use chaotic attractors from the Dysts database [26].

Under identity mixing, DCL-SINDy<sub>backprop</sub> outperforms its main competitor, AE-SINDy, on a small test suite of seven chaotic attractors across metrics that quantify both the reconstruction of the dynamical-system structure and the forecasting ability of the learned models. In addition, we evaluate DCL-SINDy<sub>lin-reg</sub> under identity mixing in Supplementary Table 7.1 and find that the regression-based variant performs well under no noise and under  $\sigma = 0.001$ , identifying more than 50 systems and more than 30 systems, respectively, out of a total of 71.

Motivated by these results, and by recent work showing that contrastive objectives can recover latent representations in which transformations act through linear maps [52], we further investigate DCL-SINDy<sub>lin-reg</sub> under linear mixing. In Fig. 4.2, we find that a simple least-squares estimator in the recovered latent space is able to identify the underlying dynamical regime under low noise. Furthermore, in Supplementary Figures 7.3, 7.4, and 7.5, we show that DCL-SINDy<sub>lin-reg</sub> matches the baseline performance of AE-SINDy under  $\sigma = 0.0$  and  $\sigma = 0.001$ . However, under  $\sigma = 0.01$ , AE-SINDy performs better in most metrics, indicating that the regression-based DCL-SINDy variant remains sensitive to noise in this setting.

### 5.2.2 Nonlinear Mixing

After establishing that DCL-SINDy<sub>lin-reg</sub> matches the baseline performance of current models under identity and low-noise linear mixing, we next address the more difficult problem of discovering dynamics after nonlinear mixing. In this setting, DCL-SINDy<sub>lin-reg</sub> significantly outperforms its main competitor, AE-SINDy, according to the metrics used in Sections 4.4 and 4.5, and recovers the Lorenz attractor nearly perfectly in Fig. 4.4.

Our results therefore suggest that AE-SINDy performs well when the representation-learning problem is simple, but becomes unstable under more complex nonlinear mixing. In contrast, DCL-SINDy<sub>lin-reg</sub> remains effective in this regime. We hypothesize that this difference arises

from three main factors: first, DCL-SINDy uses a rollout-based dynamics loss rather than relying directly on finite-difference derivative matching; second, the least-squares variant avoids sequential thresholding during encoder training; and third, the contrastive objective appears to benefit from noisy observations[53]. In the next sections, we discuss these possible explanations in more detail.

### 5.3. Why Does AE-SINDy Fail?

One possible explanation for the failure modes of AE-SINDy is its reliance on finite-difference derivative estimates. In AE-SINDy, the dynamics loss penalizes the discrepancy between the SINDy-predicted derivatives and derivative estimates computed from the observed and latent trajectories. Such finite-difference estimates are known to be sensitive to noise [54], [55]. This is also a known difficulty in SINDy, where inaccurate derivative estimation can lead to poor coefficient recovery and may cause the fitted model to overfit noise in the derivative signal [56]. The same issue motivates weak formulations of SINDy, such as W-SINDy, which replace pointwise derivative matching with integral terms and are therefore more robust under noisy observations [50], [57].

Based on our experience training AE-SINDy models, a second possible factor is the use of sequentially thresholded least squares (STLSQ) during AE-SINDy training. In AE-SINDy, coefficients of the SINDy matrix are sequentially and permanently removed once their magnitude falls below a fixed threshold [25]. After such a thresholding step, the encoder and decoder continue training under the constraint imposed by the reduced coefficient matrix. This may be beneficial when the support is identified correctly, but under complex nonlinear mixing and high noise it can also remove relevant terms too early. In this case, the encoder, decoder, and SINDy model must adapt to an incorrect sparsity structure, which may restrict optimization and prevent convergence to a satisfactory solution.

Overall, we hypothesize that AE-SINDy fails under the nonlinear-mixing regime because several difficulties compound: the derivative-based loss becomes less reliable under noisy observations, STLSQ may impose an incorrect sparsity pattern during training, and the autoencoder objective itself may become less stable in the presence of noise [42], [58]. Together, these effects provide a plausible explanation for the failure modes observed in Sections 4.4 and 4.5, and further motivate the DCL-SINDy framework proposed in this work.

## 5.4. Why Does DCL Work Under Nonlinear Mixing and Noisy Observations?

In contrast to AE-SINDy, DCL-SINDy<sub>lin-reg</sub> performs better in several metrics as the noise level increases, as shown in Fig. 4.6 and Fig. 4.5. We hypothesize that this behaviour is related to the contrastive objective. In the InfoNCE loss used in Eq. (4.1), negative samples are drawn uniformly from the dataset. This sampling strategy implicitly assumes that, for a sufficiently large batch, most randomly sampled negatives are dissimilar from the reference and positive samples. Under this condition, the contrastive loss can effectively pull the positive pair together while pushing negative samples away [30].

This assumption may be less well satisfied in the noiseless setting. When the trajectory is confined to a low-dimensional attractor, many samples may lie close to each other in state space, even if they are not temporal positives. As a result, uniformly sampled negatives may include points that are geometrically close to the reference or to the positive sample, weakening the contrastive signal. In the presence of process noise, however, trajectories are transiently perturbed away from the deterministic attractor. This can increase the spatial spread of the samples and make randomly chosen negatives more distinct from the reference-positive pair.

This effect is especially relevant for chaotic systems, where the attractor structure constrains the long-term distribution of the trajectory in phase space [59]. In such systems, moderate noise may improve the effective sampling of the local neighbourhood around the attractor and thereby make the contrastive task better conditioned. We therefore interpret the improved performance of DCL-SINDy<sub>lin-reg</sub> under noisy nonlinear mixing as a sign that the contrastive representation-learning objective utilized in the framework may require sufficient sample diversity to recover a useful latent space.

## 5.5. Current Limitations and Future Directions

### 5.5.1 Limitations

The apparent benefit of noisy observations for the contrastive representation-learning objective leads to a major limitation of the current algorithm. DCL-SINDy<sub>lin-reg</sub> relies on least-squares-based regression to avoid the thresholding problems associated with the STLSQ procedure used in AE-SINDy. However, simple LSQ methods are known to be ill-conditioned under noise [60], [61]. To address this, we implement regularization and post-training refinement in the DCL-SINDy<sub>lin-reg</sub> framework (Section 4.1). Nevertheless, as demonstrated by the higher-noise linear-

mixing results in Supplementary Figure 7.5, these additions are not yet sufficient to fully resolve the discrepancy between the noise levels that appear beneficial for the encoder and those tolerated by the dynamics-fitting step. Resolving this discrepancy is therefore a central future direction of this work.

### 5.5.2 Future Directions and Applications

Our approach to address the limitation described above is twofold. First, we aim to improve the LSQ-based dynamics-fitting step so that it becomes more robust to noise. Second, we aim to reduce the apparent dependence of the contrastive representation-learning objective on noisy observations.

To improve SINDy fitting under noise, we aim to explore recent advances in the SINDy literature. In particular, we plan to investigate weak formulations of SINDy, such as W-SINDy, which replace pointwise derivative matching with integral terms and therefore solve an integral form of the dynamics rather than directly estimating  $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, t)$ . This reduces the effect of noise during dynamics fitting because the weak formulation averages over observations and avoids explicit finite-difference derivative estimation [50], [57]. In parallel, we aim to explore alternative contrastive objectives beyond InfoNCE, such as those introduced in [62], [63], [64], to test whether the representation-learning step can be made less dependent on process noise.

Together, these extensions aim to reduce the current mismatch between the high-noise regime that appears favourable for contrastive representation learning and the lower-noise regime preferred by LSQ-based dynamics fitting. This is important because dynamical systems provide a useful perspective for analysing time-series data across many scientific domains [6], including molecular systems [7] and neuronal dynamics [8], [9]. As shown in this work, DCL-SINDy can already recover chaotic dynamics from noisy, nonlinearly mixed observations and outperforms AE-SINDy under this difficult setting. Therefore, if the remaining noise-robustness limitations can be addressed, DCL-SINDy may combine the theoretical identifiability guarantees of Dynamics Contrastive Learning [30] with the interpretability of SINDy, allowing human-readable analytical expressions to be recovered from complex measurements in a consistent manner. In this way, DCL-SINDy could contribute to the broader goal of automated knowledge discovery directly from data [1], [5], [17].

# Chapter 6

## Summary

Data-driven hypothesis discovery from time-series data requires methods that are not only predictive, but also interpretable and identifiable. This is difficult when the observed variables are high-dimensional, nonlinearly mixed measurements of a lower-dimensional dynamical system. SINDy provides a principled approach for discovering sparse analytical governing equations, but assumes that the data are already expressed in a coordinate system where the dynamics are sparse. In this thesis, we addressed this limitation by combining the identifiability-oriented representation learning of Dynamics Contrastive Learning with the interpretability of SINDy, yielding the DCL-SINDy framework.

The first contribution of the work is the implementation of DCL-SINDy as a latent-space system identification method. The framework learns an encoder that maps high-dimensional observations into a recovered latent space and fits a sparse SINDy model directly in this space. Two algorithmic variants were implemented: DCL-SINDy<sub>backprop</sub>, in which the SINDy coefficients are optimized jointly with the encoder by gradient descent, and DCL-SINDy<sub>lin-reg</sub>, in which the coefficients are re-estimated by a least-squares fit during training. To evaluate these models, we constructed a benchmark pipeline based on chaotic systems from the Dysts database, where the ground-truth latent dynamics and SINDy coefficient matrices are known. This made it possible to assess the full latent-space discovery problem at three levels: recovery of the latent representation, trajectory prediction under the learned dynamics, and reconstruction of the governing equations.

Under identity mixing, where the observed variables coincide with the latent state variables, DCL-SINDy<sub>backprop</sub> matched and, at higher noise levels, outperformed AE-SINDy in both coefficient recovery and rollout prediction. DCL-SINDy<sub>lin-reg</sub> was additionally evaluated on a larger set of chaotic attractors and recovered the correct equations for a large fraction of systems under no or low noise, showing that the framework can reproduce sparse dynamics discovery when no representation learning is required. Under linear mixing, DCL-SINDy<sub>lin-reg</sub> recovered accurate

long-horizon rollouts and the correct coefficient structure for most tested systems under low noise, establishing baseline performance in a setting where the latent space is identifiable only up to an affine transformation. The central result of the thesis was obtained under nonlinear mixing, where the Lorenz system was observed only after transformation by a linear map and an injective nonlinear MLP. In this regime,  $\text{DCL-SINDy}_{\text{lin-reg}}$  recovered the latent Lorenz attractor with high affine-aligned  $R^2$  scores across mixing depths, maintained accurate long-horizon rollout predictions, and achieved better coefficient-space recovery than AE-SINDy under the most difficult noisy nonlinear settings. Together, these results show that DCL-SINDy reaches the expected baseline performance under simpler observation regimes and substantially outperforms AE-SINDy on the selected nonlinear-mixing benchmark.

If the remaining noise-related mismatch between the encoder and dynamics-fitting modules is overcome, DCL-SINDy could become a robust framework for discovering interpretable latent dynamics from complex time-series measurements. Its broader potential lies in combining the affine identifiability guarantees of DCL with the interpretability of SINDy, allowing human-readable analytical equations to be recovered from high-dimensional, nonlinearly mixed observations in a consistent manner. This would be especially relevant in scientific domains where repeated experiments should yield comparable latent representations and governing equations, such as molecular systems and neuronal dynamics. In this sense, DCL-SINDy represents a promising step toward repeatable, interpretable, and data-driven discovery of latent dynamical laws directly from observed time-series data.

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# Chapter 7

## Supplementary Material

### 7.1. Tested Attractors

The experiments are conducted on the following 12 chaotic attractors drawn from the dysts database:

| Lorenz                        | SprottTorus               | VallisElNino           | Dadras                      |       |
|-------------------------------|---------------------------|------------------------|-----------------------------|-------|
| $\dot{x} = 10(y - x)$         | $\dot{x} = y + 2xy + xz$  | $\dot{x} = -3x + 102y$ | $\dot{x} = -3x + y + 2.7yz$ | (7.1) |
| $\dot{y} = 28x - y - xz$      | $\dot{y} = 1 - 2x^2 + yz$ | $\dot{y} = -y + xz$    | $\dot{y} = 1.7y + z - xz$   |       |
| $\dot{z} = xy - \frac{8}{3}z$ | $\dot{z} = x - x^2 - y^2$ | $\dot{z} = 1 - z - xy$ | $\dot{z} = 2xy - 9z$        |       |

| Lorenz84                               | PehlivanWei              | NoseHoover            | SprottA             |       |
|----------------------------------------|--------------------------|-----------------------|---------------------|-------|
| $\dot{x} = 1.32(4.83 - x) - y^2 - z^2$ | $\dot{x} = y - yz$       | $\dot{x} = y$         | $\dot{x} = y$       | (7.2) |
| $\dot{y} = 4.194 - y + xy - 7.91xz$    | $\dot{y} = -2x + y + yz$ | $\dot{y} = -x + yz$   | $\dot{y} = -x + yz$ |       |
| $\dot{z} = -z + 7.91xy + xz$           | $\dot{z} = 2 - xy - y^2$ | $\dot{z} = 1.5 - y^2$ | $\dot{z} = 1 - y^2$ |       |

| SprottB            | RikitakeDynamo          | Rucklidge                   | Sakarya                    |       |
|--------------------|-------------------------|-----------------------------|----------------------------|-------|
| $\dot{x} = yz$     | $\dot{x} = -x + yz$     | $\dot{x} = -2x + 6.7y - yz$ | $\dot{x} = -x + y + yz$    | (7.3) |
| $\dot{y} = x - y$  | $\dot{y} = -x - y + xz$ | $\dot{y} = x$               | $\dot{y} = -x - y + 0.4xz$ |       |
| $\dot{z} = 1 - xy$ | $\dot{z} = 1 - xy$      | $\dot{z} = -z + y^2$        | $\dot{z} = z - 0.3xy$      |       |

## 7.2. DCL-SINDy<sub>lin-reg</sub> Performance on the Full Set of Chaotic Attractors

Table 7.1: Summary across all 71 attractors of the Dysts database [47] grouped by data dynamics noise level and number of initial conditions ( $\#IC$ ) and trajectory lengths ( $\#points$ ) in the training data. Each IC and points combination results in a total of 100,000 datapoints, but the length of each individual rollout is different. For example, for  $4 \times 25,000$ , 4 initial conditions are used to simulate 25,000-step long trajectories, while for  $20 \times 5,000$ , 20 ICs are used to sample 5000 points. The table reports the number of attractors (out of  $n_{\text{attractors}} = 71$ ) for which all three seeds recovered a zero-Hamming-distance model, together with the mean Frobenius coefficient error and mean prediction  $R^2$  computed only over these zero-Hamming-distance cases.

| Noise std.<br>( $\sigma$ ) | $\#IC \times \#points$<br>( $\#points_{\text{total}} = 100,000$ ) | Zero-Hamming<br>attractors<br>(3/3 seeds) | Mean Frobenius<br>error<br>( $H = 0$ ) | Mean<br>Predictive $R^2$<br>( $H = 0$ ) |
|----------------------------|-------------------------------------------------------------------|-------------------------------------------|----------------------------------------|-----------------------------------------|
| 0.000                      | $4 \times 25,000$                                                 | 54                                        | $2.0 \times 10^{-4}$                   | 1.0000                                  |
|                            | $20 \times 5000$                                                  | 53                                        | $2.0 \times 10^{-4}$                   | 1.0000                                  |
|                            | $100 \times 1000$                                                 | 61                                        | $9.0 \times 10^{-4}$                   | 1.0000                                  |
| 0.001                      | $4 \times 25,000$                                                 | 35                                        | $9.9 \times 10^{-3}$                   | 1.0000                                  |
|                            | $20 \times 5000$                                                  | 40                                        | $1.48 \times 10^{-2}$                  | 1.0000                                  |
|                            | $100 \times 1000$                                                 | 33                                        | $1.56 \times 10^{-2}$                  | 1.0000                                  |
| 0.010                      | $4 \times 25,000$                                                 | 8                                         | $6.89 \times 10^{-2}$                  | 0.9999                                  |
|                            | $20 \times 5000$                                                  | 8                                         | $6.54 \times 10^{-2}$                  | 0.9999                                  |
|                            | $100 \times 1000$                                                 | 3                                         | $1.013 \times 10^{-1}$                 | 0.9996                                  |
| 0.100                      | $4 \times 25,000$                                                 | 0                                         | –                                      | –                                       |
|                            | $20 \times 5000$                                                  | 0                                         | –                                      | –                                       |
|                            | $100 \times 1000$                                                 | 0                                         | –                                      | –                                       |

### 7.3. Predictive Rollouts after Identity Mixing

#### DCL-SINDy<sub>backprop</sub> Rollouts under Identity Mixing

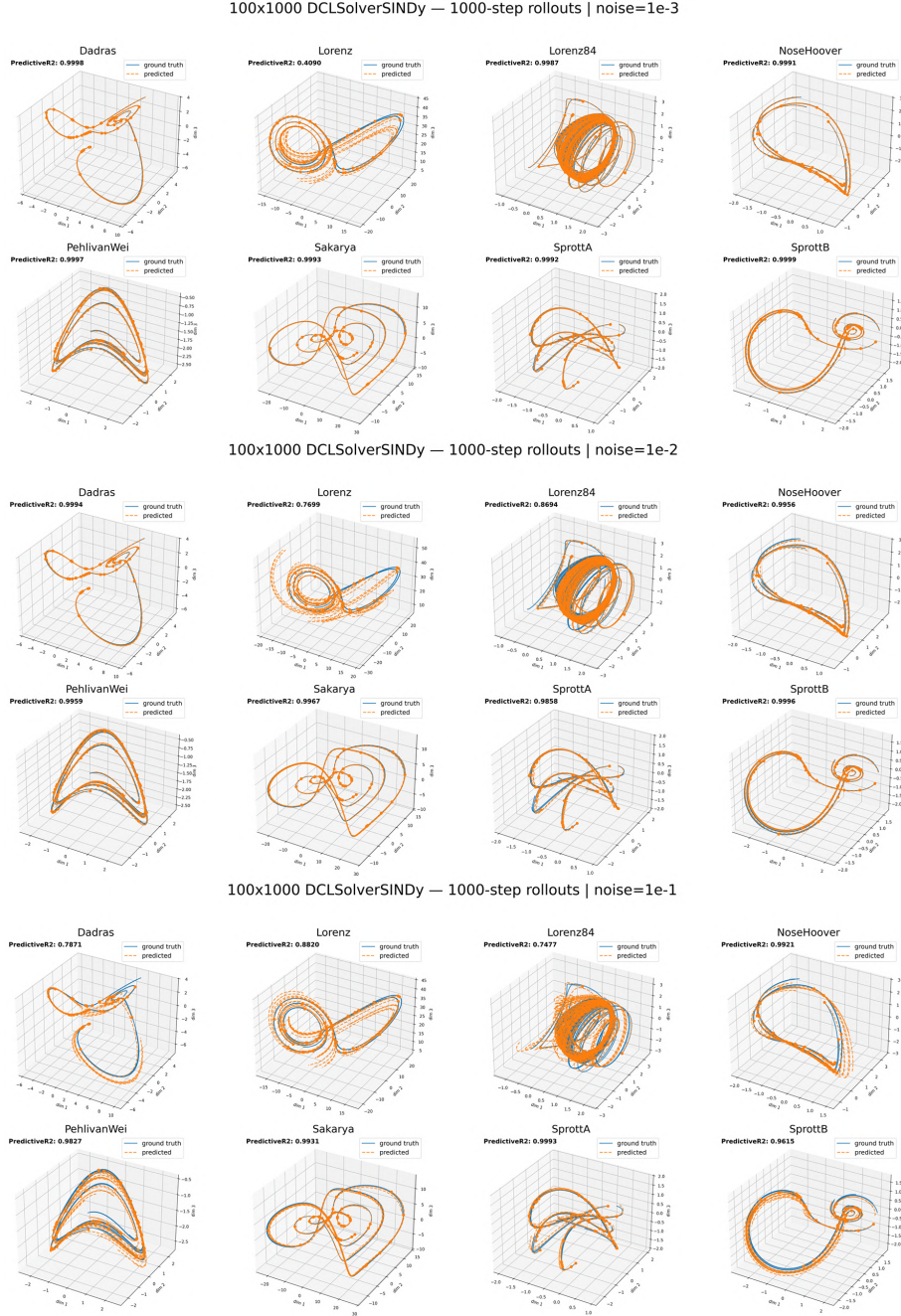
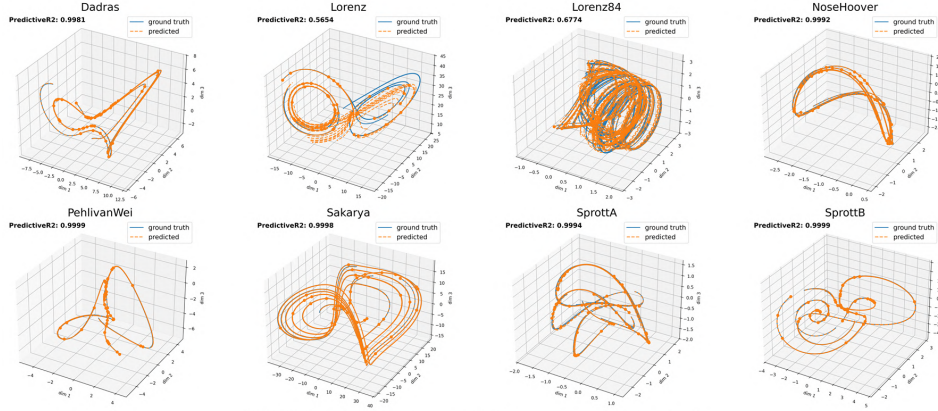


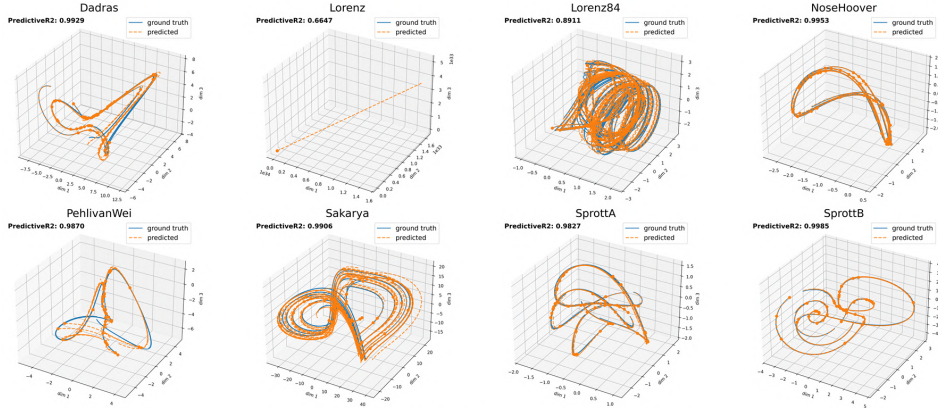
Figure 7.1: Extended 1000-step rollout predictions of DCL-SINDy<sub>backprop</sub> under identity mixing (Section 3.3.2) at process-noise levels  $\sigma = 10^{-3}$ ,  $\sigma = 10^{-2}$ , and  $\sigma = 10^{-1}$  (top to bottom). Seven of the shown attractors correspond to the identity-mixing benchmark from Section 4.2. Lorenz is additionally included only for visual completeness of this supplementary rollout figure. Each subpanel is a 3D phase portrait of one attractor, with the ground-truth test trajectory in blue and the model rollout in orange; the title reports the predictive  $R^2$  at the 1000-step horizon (Section 3.2.4, Eq. (3.16)). DCL-SINDy<sub>backprop</sub> retains high rollout accuracy across all noise levels, complementing the 750-step comparison in Figure 4.1c.

## AE-SINDy Rollouts under Identity Mixing

100x1000 AESINDySolver — 1000-step rollouts | noise=1e-3



100x1000 AESINDySolver — 1000-step rollouts | noise=1e-2



100x1000 AESINDySolver — 1000-step rollouts | noise=1e-1

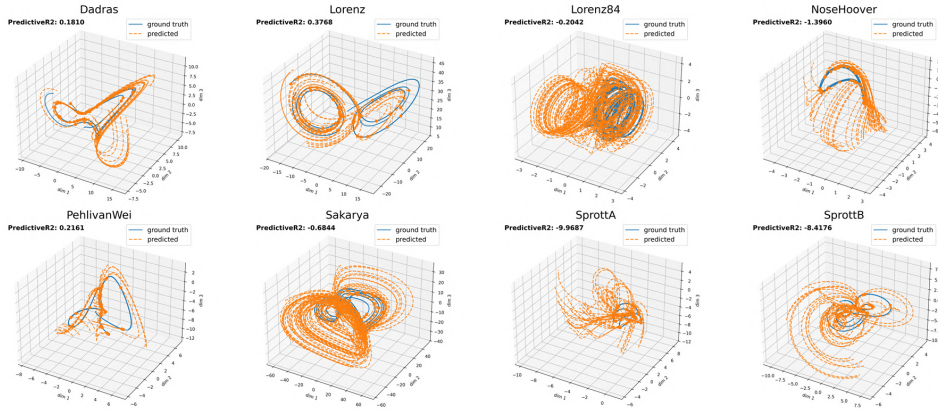


Figure 7.2: Extended 1000-step rollout predictions of AE-SINDy under identity mixing (Section 3.3.2) at process-noise levels  $\sigma = 10^{-3}$ ,  $\sigma = 10^{-2}$ , and  $\sigma = 10^{-1}$  (top to bottom). Seven of the shown attractors correspond to the identity-mixing benchmark from Section 4.2. Lorenz is additionally included only for visual completeness of this supplementary rollout figure. Each subpanel is a 3D phase portrait of one attractor, with the ground-truth test trajectory in blue and the model rollout in orange; the title reports the predictive  $R^2$  at the 1000-step horizon (Section 3.2.4, Eq. (3.16)). Compared with DCL-SINDy<sub>backprop</sub> in Figure 7.1, AE-SINDy degrades markedly as the noise level increases, most visibly at  $\sigma = 10^{-1}$ , where several rollouts diverge and the predictive  $R^2$  drops or becomes negative.

## 7.4. Linear Mixing Statistics

Linear Mixing Statistics at  $\sigma = 0.0$

|                | AE-SINDy |                                   |          |          | $\sigma = 0.0$ | DCL-SINDy |                                   |          |          |
|----------------|----------|-----------------------------------|----------|----------|----------------|-----------|-----------------------------------|----------|----------|
|                | MCC      | iDynR <sub>500</sub> <sup>2</sup> | iFroNorm | iHamming |                | MCC       | iDynR <sub>500</sub> <sup>2</sup> | iFroNorm | iHamming |
| Dadras         | 7.57e-01 | 1.00e+00                          | 4.88e-04 | 4.44e-02 |                | 7.35e-01  | 1.00e+00                          | 1.45e-02 | 0.00e+00 |
| Lorenz         | 8.39e-01 | 9.99e-01                          | 7.13e-01 | 0.00e+00 |                | 8.96e-01  | 1.00e+00                          | 1.19e-02 | 0.00e+00 |
| Lorenz84       | 8.42e-01 | 1.00e+00                          | 1.10e-03 | 0.00e+00 |                | 8.69e-01  | 9.68e-01                          | 6.83e-03 | 0.00e+00 |
| NoseHoover     | 8.17e-01 | 1.00e+00                          | 2.80e-05 | 0.00e+00 |                | 8.67e-01  | 1.00e+00                          | 2.85e-04 | 0.00e+00 |
| PehlivanWei    | 8.58e-01 | 1.00e+00                          | 2.15e-04 | 0.00e+00 |                | 8.85e-01  | 1.00e+00                          | 9.43e-04 | 0.00e+00 |
| RikitakeDynamo | 8.73e-01 | 1.00e+00                          | 2.82e-03 | 0.00e+00 |                | 9.51e-01  | 1.00e+00                          | 3.38e-03 | 0.00e+00 |
| Rucklidge      | 7.59e-01 | 1.00e+00                          | 1.28e-02 | 0.00e+00 |                | 8.27e-01  | 1.00e+00                          | 1.24e-03 | 0.00e+00 |
| Sakarya        | 7.25e-01 | 1.00e+00                          | 9.90e-04 | 0.00e+00 |                | 7.95e-01  | 1.00e+00                          | 7.46e-02 | 0.00e+00 |
| SprottA        | 7.74e-01 | 1.00e+00                          | 7.06e-05 | 0.00e+00 |                | 8.19e-01  | 1.00e+00                          | 3.01e-04 | 0.00e+00 |
| SprottB        | 8.80e-01 | 1.00e+00                          | 4.81e-05 | 0.00e+00 |                | 8.96e-01  | 1.00e+00                          | 1.07e-03 | 0.00e+00 |
| SprottTorus    | 7.75e-01 | 1.00e+00                          | 1.16e-04 | 0.00e+00 |                | 7.95e-01  | 1.00e+00                          | 2.20e-03 | 0.00e+00 |
| VallisElNino   | 6.25e-01 | 1.00e+00                          | 1.98e-02 | 0.00e+00 |                | 5.83e-01  | 1.00e+00                          | 8.99e-04 | 0.00e+00 |

Figure 7.3: Full quantitative comparison of AE-SINDy (left block) and DCL-SINDy<sub>lin-reg</sub> (right block) across all twelve benchmark attractors under linear mixing (Section 4.3) at process-noise level  $\sigma = 0.0$ . The rows of the heatmap table are the twelve attractors and the columns report the MCC (Section 3.2.1), the identifiable rollout score iDynR<sub>500</sub><sup>2</sup> (Section 3.2.4), the normalized identifiable Frobenius coefficient error iFroNorm, and the signed identifiable Hamming distance iHamming (Section 3.2.3), all reported in their *Identifiable* form (Sections 3.2.1 and 3.2.2). Greener cells indicate better scores (higher MCC and iDynR<sup>2</sup>, lower iFroNorm and iHamming). At  $\sigma = 0.0$  the two methods are closely matched, both achieving near-zero Hamming distance and high rollout scores.

## Linear Mixing Statistics at $\sigma = 0.001$

|                | AE-SINDy |                                   |          |          | $\sigma = 0.001$ | DCL-SINDy |                                   |          |          |
|----------------|----------|-----------------------------------|----------|----------|------------------|-----------|-----------------------------------|----------|----------|
|                | MCC      | iDynR <sub>500</sub> <sup>2</sup> | iFroNorm | iHamming |                  | MCC       | iDynR <sub>500</sub> <sup>2</sup> | iFroNorm | iHamming |
| Dadras         | 7.65e-01 | 1.00e+00                          | 2.16e-03 | 1.11e-02 |                  | 7.37e-01  | 9.97e-01                          | 1.08e-01 | 0.00e+00 |
| Lorenz         | 8.50e-01 | 9.99e-01                          | 8.02e-01 | 0.00e+00 |                  | 8.96e-01  | 1.00e+00                          | 1.81e-01 | 0.00e+00 |
| Lorenz84       | 8.38e-01 | 1.00e+00                          | 6.13e-03 | 0.00e+00 |                  | 8.77e-01  | 9.75e-01                          | 1.60e-02 | 0.00e+00 |
| NoseHoover     | 9.19e-01 | 1.00e+00                          | 5.44e-03 | 1.11e-02 |                  | 8.69e-01  | 9.98e-01                          | 6.43e-02 | 0.00e+00 |
| PehlivanWei    | 8.47e-01 | 1.00e+00                          | 3.81e-03 | 0.00e+00 |                  | 8.82e-01  | 9.99e-01                          | 5.68e-02 | 0.00e+00 |
| RikitakeDynamo | 8.62e-01 | 1.00e+00                          | 4.70e-02 | 0.00e+00 |                  | 9.46e-01  | 9.98e-01                          | 2.73e-01 | 0.00e+00 |
| Rucklidge      | 7.19e-01 | 1.00e+00                          | 3.69e-03 | 0.00e+00 |                  | 8.42e-01  | 1.00e+00                          | 2.80e-01 | 0.00e+00 |
| Sakarya        | 7.25e-01 | 1.00e+00                          | 9.45e-03 | 0.00e+00 |                  | 7.93e-01  | 1.00e+00                          | 4.02e-01 | 0.00e+00 |
| SprottA        | 7.71e-01 | 1.00e+00                          | 4.48e-03 | 0.00e+00 |                  | 8.22e-01  | 9.96e-01                          | 5.63e-02 | 0.00e+00 |
| SprottB        | 8.82e-01 | 1.00e+00                          | 3.68e-03 | 2.22e-02 |                  | 8.80e-01  | 9.98e-01                          | 3.26e-02 | 0.00e+00 |
| SprottTorus    | 7.64e-01 | 1.00e+00                          | 3.32e-03 | 0.00e+00 |                  | 8.07e-01  | 9.89e-01                          | 4.57e-02 | 0.00e+00 |
| VallisElNino   | 6.25e-01 | 1.00e+00                          | 4.55e-02 | 0.00e+00 |                  | 5.79e-01  | 9.94e-01                          | 1.94e-02 | 0.00e+00 |

Figure 7.4: Full quantitative comparison of AE-SINDy (left block) and DCL-SINDy<sub>lin-reg</sub> (right block) across all twelve benchmark attractors under linear mixing (Section 4.3) at process-noise level  $\sigma = 0.001$ . The rows of the heatmap table are the twelve attractors and the columns report the MCC (Section 3.2.1), the identifiable rollout score iDynR<sub>500</sub><sup>2</sup> (Section 3.2.4), the normalized identifiable Frobenius coefficient error iFroNorm, and the signed identifiable Hamming distance iHamming (Section 3.2.3), all reported in their *Identifiable* form (Sections 3.2.1 and 3.2.2). Greener cells indicate better scores (higher MCC and iDynR<sup>2</sup>, lower iFroNorm and iHamming). At  $\sigma = 0.001$  the two methods remain closely matched, with both still recovering the correct support (near-zero Hamming distance) for most attractors.

## Linear Mixing Statistics at $\sigma = 0.01$

|                | AE-SINDy |                                   |          |          | $\sigma = 0.01$ | DCL-SINDy |                                   |          |          |
|----------------|----------|-----------------------------------|----------|----------|-----------------|-----------|-----------------------------------|----------|----------|
|                | MCC      | iDynR <sub>500</sub> <sup>2</sup> | iFroNorm | iHamming |                 | MCC       | iDynR <sub>500</sub> <sup>2</sup> | iFroNorm | iHamming |
| Dadras         | 7.70e-01 | 9.99e-01                          | 1.97e-02 | 1.11e-02 |                 | 7.68e-01  | 9.36e-01                          | 6.17e-01 | 0.00e+00 |
| Lorenz         | 8.53e-01 | 9.98e-01                          | 1.28e+00 | 0.00e+00 |                 | 9.11e-01  | 9.98e-01                          | 9.26e-01 | 0.00e+00 |
| Lorenz84       | 7.18e-01 | 9.94e-01                          | 2.06e-02 | 0.00e+00 |                 | 8.77e-01  | 5.97e-01                          | 1.26e-01 | 0.00e+00 |
| NoseHoover     | 1.00e+00 | 1.00e+00                          | 2.59e-02 | 2.11e-01 |                 | 8.76e-01  | 8.16e-01                          | 4.08e-01 | 0.00e+00 |
| PehlivanWei    | 7.62e-01 | 1.00e+00                          | 2.69e-02 | 0.00e+00 |                 | 8.85e-01  | 9.42e-01                          | 3.49e-01 | 0.00e+00 |
| RikitakeDynamo | 8.67e-01 | 9.98e-01                          | 3.04e-01 | 0.00e+00 |                 | 9.07e-01  | 8.22e-01                          | 7.03e-01 | 0.00e+00 |
| Rucklidge      | 7.61e-01 | 9.99e-01                          | 1.38e-01 | 0.00e+00 |                 | 9.01e-01  | 9.22e-01                          | 8.56e-01 | 0.00e+00 |
| Sakarya        | 7.29e-01 | 1.00e+00                          | 1.94e-01 | 0.00e+00 |                 | 7.70e-01  | 9.87e-01                          | 9.61e-01 | 0.00e+00 |
| SprottA        | 9.92e-01 | 9.99e-01                          | 4.37e-02 | 1.44e-01 |                 | 8.85e-01  | 7.02e-01                          | 4.26e-01 | 0.00e+00 |
| SprottB        | 8.44e-01 | 9.99e-01                          | 4.22e-02 | 1.00e-01 |                 | 8.12e-01  | 7.22e-01                          | 2.11e-01 | 0.00e+00 |
| SprottTorus    | 8.16e-01 | 9.81e-01                          | 7.93e-02 | 0.00e+00 |                 | 7.77e-01  | 5.32e-01                          | 4.39e-01 | 0.00e+00 |
| VallisElNino   | 6.09e-01 | 9.79e-01                          | 1.93e-01 | 0.00e+00 |                 | 5.77e-01  | 7.33e-01                          | 2.20e-01 | 0.00e+00 |

Figure 7.5: Full quantitative comparison of AE-SINDy (left block) and DCL-SINDy<sub>lin-reg</sub> (right block) across all twelve benchmark attractors under linear mixing (Section 4.3) at process-noise level  $\sigma = 0.01$ . The rows of the heatmap table are the twelve attractors and the columns report the MCC (Section 3.2.1), the identifiable rollout score iDynR<sub>500</sub><sup>2</sup> (Section 3.2.4), the normalized identifiable Frobenius coefficient error iFroNorm, and the signed identifiable Hamming distance iHamming (Section 3.2.3), all reported in their *Identifiable* form (Sections 3.2.1 and 3.2.2). Greener cells indicate better scores (higher MCC and iDynR<sup>2</sup>, lower iFroNorm and iHamming). At this higher noise level, AE-SINDy outperforms DCL-SINDy<sub>lin-reg</sub> on most metrics, motivating the move to the more challenging nonlinear-mixing regime studied in Section 4.4.

## 7.5. Divergent AE-SINDy Rollouts Under Nonlinear Mixing

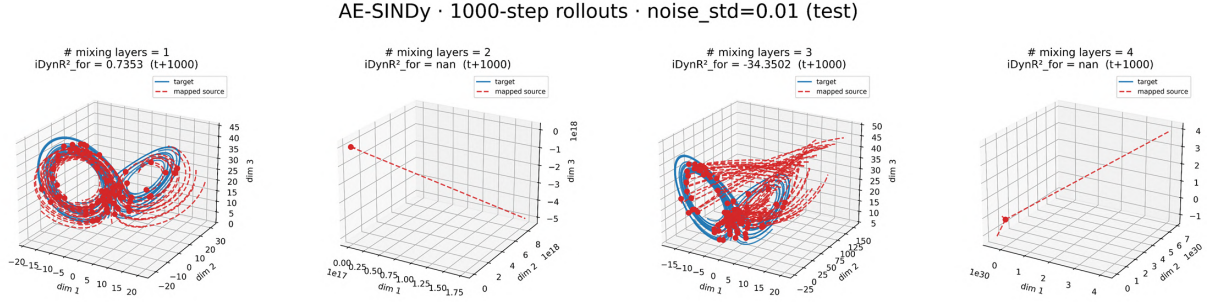


Figure 7.6: Numerically unstable AE-SINDy 1000-step rollouts on the Lorenz system under noisy nonlinear mixing, illustrating the divergent and NaN-valued behaviour referenced in Section 4.4 and Figure 4.4. The data are generated with the leaky-ReLU MLP mixing function  $\mathbf{g}_{\text{ln}}(\mathbf{x}) = \text{MLP}_l(\mathbf{A}\mathbf{x})$  (Section 3.1.2) at the high-noise level  $\sigma = 0.01$ , with the number of mixing layers (here # mixing layers) increasing from  $l = 1$  to  $l = 4$  across the four panels (left to right). Each panel shows a 1000-step rollout phase portrait, with the ground-truth target trajectory in blue and the affine-aligned AE-SINDy rollout (“mapped source”) as a red dashed curve, and the title reports the forward-biased identifiable dynamics score  $\text{iDynR}_{\text{fwd}}^2$  at the 1000-step horizon (Section 3.2.4, Eq. (3.16)). The figure shows a representative AE-SINDy model selected on the validation set: at  $l = 1$  the rollout still loosely tracks the attractor ( $\text{iDynR}_{\text{fwd}}^2 = 0.7353$ ), but for  $l = 2$  and  $l = 4$  the integrated trajectory diverges to scales of order  $10^{17}$ – $10^{30}$  and the score evaluates to NaN, while at  $l = 3$  it collapses to a large negative value ( $\text{iDynR}_{\text{fwd}}^2 = -34.35$ ). This explains why, in the main-text comparison of Figure 4.4, only one out of six trained AE-SINDy models yielded finite  $\text{iDynR}^2$  values at  $l = 4$  and  $\sigma = 0.01$ , and motivates the stability analysis in the ablation study (Section 4.5).